Tax Incidence with Endogenous Exit

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I show that measured tax incidence is biased when agents differentially exit the observed tax base. The bias depends on the magnitude of exit and the sign of the covariance between exit and incidence. These together provide a sufficient-statistic correction to measured incidence and the marginal value of public funds. The correction requires no identification of who exits or why and applies wherever heterogeneous agents can exit a tax base. Applied to the 2018–19 trade war between the United States and China, the correction places a lower bound on domestic incidence of 0.86, against the literature benchmark of full passthrough, which remains the upper bound in my framework.

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1 Introduction

Canonically, tax incidence is governed by the relative elasticities of supply and demand and market structure (Gruber 2019). However, two common empirical features make it difficult to assess tax incidence following a reform. First, agents typically differ in how much incidence they bear. For example, when a tax falls on firms with heterogeneous market power, they will systematically differ in how much of the tax they can pass on to consumers. Second, agents typically have the option to exit the observed tax base through, for example, avoidance, evasion, or shutdown. When the two features covary, estimated tax incidence reflects survivorship bias by conditioning on the agents who remain, rather than the original population. That bias has two consequences: measured incidence is wrong, and welfare evaluations of tax instruments inherit the bias. This paper provides a sufficient statistics approach to bounding the bias and applies it to the 2018 U.S.-China trade war.

Two sufficient statistics summarize the bias. The first is the share of the original tax base that exits following the reform. The second is the direction of selection: whether the agents who leave bear systematically more or less incidence than those who stay. While the first is generally observable, the second typically requires outside information or theoretical input. For example, when firms have labor market power, an income tax hike falls on workers; those bearing the heaviest burden are most likely to exit to the gray market. We would have to know something about the market and the consequences of monopsony to sign the covariance between exit and incidence. Nevertheless, the framework's agnosticism leaves the direction an empirical question rather than a maintained assumption.

Magnitude and direction are sufficient to place worst-case bounds on the degree of bias. With magnitude, we can concretely scale the width of the bound, while direction guarantees that measured incidence is a lower or upper bound. When the covariance between exit and incidence is positive, measured incidence is an upper bound, and vice versa when the covariance is negative. Importantly, there may be multiple types of exit pulling in opposite directions. The framework accommodates that reality and shows how to handle it empirically. Moreover, the bounds can in principle be tightened by recovering the passthrough of the marginal exiter—though empirical implementation is data-demanding. Altogether, the results adapt the worst-case selection bounds of Manski (1989) and Manski and Pepper (2003) to a survivor–cohort setting.

Biased incidence propagates into biased welfare measurement. The marginal value of public funds (MVPF) measures the welfare cost of raising a marginal dollar via a particular tax instrument (Mayshar 1990; Bastani 2025). The MVPF's numerator is the planner-weighted incidence on each side of the market. When the planner weights the two sides asymmetrically, biased incidence biases the numerator. For example, a nationalist planner would only value domestic welfare, so if measured tariff incidence is biased, so is the numerator of the MVPF. The denominator captures the fiscal externality from exit. It shrinks as the tax elasticity of exit rises. I show that three empirical inputs bound welfare under selective exit: the exit share, the sign of selection, and the exit elasticity. I provide all three for the 2018 U.S.-China trade war.

I apply the framework to the 2018 U.S.-China trade war. A large literature using bilateral customs data finds passthrough near one, implying that U.S. importers bore the full burden of tariffs (Amiti, Redding, and Weinstein 2019, 2020; Fajgelbaum et al. 2020; Cavallo et al. 2021). This presents a puzzle because, in theory, terms-of-trade effects would imply that China should have shared some of the burden. Because customs data condition on direct shipments, they identify the average passthrough among survivors rather than the pre-tariff cohort. Two plausible exit margins may influence the direction of bias in a way that depends on theory. First, a growing literature establishes that Chinese exporters responded to tariffs by shipping through third countries (Iyoha et al. 2025; Freund 2024). This *transshipment* allows exporters to evade tariffs and hides them from bilateral customs data. Second, standard trade models predict that higher tariffs cause the least profitable exporters to shut down, in the spirit of Melitz (2003). To help sign the covariance, I formalize the two margins of exit in a partial equilibrium trade model. Exporters face Kimball (1995) demand from U.S. consumers and choose among three options: ship directly, transship, or shut down. Kimball demand, empirically realistic and standard (Gopinath and Itskhoki 2010; De Loecker, Eeckhout, and Unger 2020; Edmond, Midrigan, and Xu 2023), makes the elasticity of demand rise with price. Markups therefore vary with cost: low-cost exporters earn deep markups, while high-cost exporters operate at thin ones. When tariffs rise, low-cost exporters absorb the shock by compressing their markups; high-cost exporters have no markup to give and pass tariffs through. Cost therefore determines incidence. Exporters who would otherwise bear the most of the tariff transship, while those who would bear the least shut down. The two margins push the bias in opposite directions, and the puzzle's resolution turns on the overall magnitude of exit and the sign of selection.

To measure the two margins and maintain comparability with the passthrough literature, I extend Freund (2024) to come up with two sets of rigorous screening criteria to identify transshipment and shutdown using bilateral customs data. While few studies directly measure either margin with granular data, I do find that my screens closely match the firm-level evidence from Iyoha et al. (2025). Overall, I find that by 2023, transshipment reaches 5–14% of baseline China–U.S. trade, concentrated in intermediates (7–21%) and capital (6–18%). By comparison, shutdown is negligible in the aggregate and is only comparable to transshipment for consumption goods, for which both transshipment and shutdown are around 2%. In other words, transshipment dominates shutdown by a considerable margin, so theory predicts a positive covariance between exit and passthrough. To validate the theory, I collect three pieces of evidence. First, I show that survivor passthrough rises with exit intensity, and this is only true within differentiated goods. That matters because it suggests that market power is mediating passthrough, just as Kimball demand suggests. Second, I show that the exit response is decreasing in third country tariffs, confirming selection operates through the transshipment mechanism. Third, I rule out a compositional-shift alternative: within-cell unit values compress from the bottom of the distribution, not the top. Altogether, theory and evidence agree that the covariance is positive, which allows us to place bounds on incidence and welfare.

By placing assumptions on the passthrough of exiters, we can trace the width of the bounds on both incidence and the exit-based MVPF for tariffs. Assuming exiters have zero passthrough, the aggregate corrected passthrough has a lower bound of 0.86, against a survivor-benchmark upper bound of one. At the category level, the lower bounds for capital and intermediate goods are roughly 0.80, while there is essentially no correction for consumption goods. This accords with firm-to-firm evidence from Flaaen, Hortaçsu, and Tintelnot (2020) and Cavallo et al. (2021) on consumption goods. When mapped into welfare, I find an aggregate MVPF range of roughly 1.03–1.52, with correspondingly larger upper bounds for intermediate and capital goods. Applying the incidence correction to the literature range of 1.2–1.6 (Finkelstein and Hendren 2020) drops the lower bound toward one, leaving a range broadly comparable to my exit-only estimate.

While the framework is sufficiently general, the application focuses on two exit margins and therefore yields a conservative estimate of exit. For example, firms also misreport value (Fajgelbaum and Khandelwal 2024), break up shipments, or misclassify goods (Fisman and Wei 2004), all of which are forms of avoidance. On the welfare front, the analysis is partial equilibrium and adopts a strictly nationalist perspective. It abstracts from general equilibrium effects on terms of trade and the further passthrough of costs from importers to final consumers (Amiti, Itskhoki, and Konings 2019; Flaaen et al. 2025). Despite these caveats, the framework's correction generalizes well beyond the tariff application: wherever heterogeneous agents can exit a tax base, the same sufficient statistic bounds the bias.

Related Literature. This paper relates to three strands of literature.

First, I contribute to a literature using sufficient statistics for welfare analysis. Feldstein (1999) established the elasticity of taxable income as the sufficient statistic for deadweight loss. Chetty (2009a) showed that the same elasticity overstates deadweight loss when avoidance involves transfers rather than real resource costs. Weyl and Fabinger (2013) established the parallel for incidence: passthrough is the sufficient statistic under general forms of competition. I show that selection on exit biases this statistic for the cohort passthrough that welfare requires. The exit share is a sufficient statistic for the magnitude of the gap between survivor and cohort passthrough.

The sign of the covariance between passthrough and survival determines its direction.[1] To develop these statistics, I build on Manski (1989) and Manski and Pepper (2003) to obtain worst-case bounds, which I then tighten.

Second, this paper builds on methodologies for inferring illicit trade from official statistics. Fisman and Wei (2004) introduced the trade discrepancy approach, comparing exporter and importer reports to quantify evasion; Javorcik and Narciso (2008) extended this to show that evasion concentrates in differentiated products. Fisman, Moustakerski, and Wei (2008) then applied the logic to indirect trade, finding that flows via Hong Kong rise with Chinese tariffs. My approach extends the screening methodology of Freund (2024), who flags products with trade patterns consistent with transshipment. My contribution is to add a second set of tightening screens as well as a shutdown margin. Importantly, my screens are consistent with more granular evidence from Iyoha et al. (2025) on transshipment through Vietnam. What this activity reflects, however, is contested. Alfaro and Chor (2025) note that China's value added in third-country exports remains substantial, raising the possibility that hub-to-U.S. surges reflect supply chain integration rather than pure transshipment. My screens isolate the latter by requiring China-to-hub flows to scale with the hub-to-U.S. surge: were the hub genuinely producing rather than re-exporting Chinese goods, Chinese shipments to it would not track the surge.

Third, this paper reconciles findings from the tariff incidence literature. Studies using bilateral customs data classified by country of origin find near-complete passthrough of the 2018 tariffs (Amiti, Redding, and Weinstein 2019, 2020; Fajgelbaum et al. 2020). These estimates correctly identify passthrough among firms that continue shipping directly, but when firms reroute, they exit the sample. Cavallo et al. (2021) use BLS microdata that tracks identical goods over time and find near-complete passthrough at the border; their sample weights toward consumer goods, where I find essentially no transshipment, so their estimates are consistent with my framework. Flaaen, Hortaçsu, and Tintelnot (2020) study washing machines and find that country-specific tariffs allowed firms to relocate, muting price effects, while a global tariff closing this option caused passthrough to exceed 100 percent, illustrating that measured passthrough depends on whether the avoidance margin is open, the same dependence my selection mechanism formalizes. Alviarez et al. (2023) use U.S. firm-to-firm transaction data on repeated, arm's-length intermediate-input imports and find within-match pass-through of the 2018 tariffs of 65–71 percent, substantially below product-level estimates. They attribute the gap to two-sided market power. Importers' buyer power compels exporters with decreasing returns to absorb tariffs through marginal cost adjustments, with markup channels playing a smaller, offsetting role. Compositional shifts toward one-off transactions, which exhibit higher passthrough, further inflate product-level aggre-

1. As I discuss in Section 2.3, Chetty's resource-vs-transfer distinction does not bear on the exit-based MVPF I develop.

gates. Their evidence isolates passthrough heterogeneity within direct customs shipments; the selection bias I formalize is across customs shipments, between exporters who continue direct shipping and those who reroute. My contribution is to formalize the selection mechanism, show that the bias is concentrated in business inputs where transshipment is prevalent, and derive the sufficient-statistic correction for welfare analysis.

2 Tax Incidence under Heterogeneity and Exit

In applied work, the passthrough of a tax onto consumer prices is routinely recovered by regressing prices on tax changes. Since passthrough determines the first-order incidence of the tax, the regression bears important information about welfare and the distributional burden of taxation. However, three empirically relevant features complicate the interpretation of the passthrough coefficient. First, agents within a given tax base respond differentially to tax changes. The theoretical passthrough of a corporate tax to consumer prices, for example, varies across industries depending on a firm's market power. Second, agents typically have the option to exit the observed tax base through a variety of means, including avoidance, evasion, and ceasing operation entirely. Third, the propensity to exit is generally correlated with the pricing response. Highly profitable multinationals with deep markups may absorb a tax hike rather than raise consumer prices, but they are also exactly the firms with the resources and incentives to route their profits through offshore tax havens, removing themselves from the observed domestic tax data. This section formalizes the resulting bias in estimated passthrough and bounds it using two sufficient statistics: the share of the market which exits and a sign restriction on the exit share's covariance with passthrough.

2.1 The Selection Wedge

This subsection traces measurement bias in the passthrough coefficient from a single market through to aggregate selection. The starting point is a single submarket subject to a small tax change. Under standard assumptions on supply and demand, the tax shifts the equilibrium buyer price by the passthrough coefficient β, defined as the elasticity of the buyer price with respect to the tax wedge. Under perfect competition with a small tax, β is determined entirely by the relative magnitudes of the supply and demand elasticities, $\beta = \eta_S/(\eta_S + |\eta_D|) \in [0, 1]$ (Gruber 2019). Other market structures generalize this expression. Under oligopolistic competition, for example, β depends additionally on the curvature of demand and need not lie below one.

In practice, however, tax instruments rarely apply to a single, well-defined market. An import tariff, for example, may cover hundreds of differentiated products. Submarkets within a tax base typically differ in technology, market structure, and other primitives that load onto the curvature of supply and demand. For example, in tariff applications, submarkets may differ in foreign exporter market power, which results in systematic variation across products in how easily exporters can pass the tax onto domestic prices. As a result, estimated passthrough is typically a weighted average of these features across submarkets.

To capture this heterogeneity, consider submarkets indexed by $i \in \mathcal{I}$, distributed by a measure G. Concretely, a submarket might be a foreign exporter shipping a single variety, a firm operating in a single jurisdiction, or a product line within a firm. What unites these interpretations is that a submarket is the unit at which the passthrough coefficient may vary. Each submarket i accordingly has its own local supply $S_i(p_i)$ and demand $D_i(p_i)$ parameterized by an underlying type ξ_i, yielding a submarket-specific passthrough coefficient

$$\beta_i \equiv \beta(\xi_i) \in [\beta_{\min}, \beta_{\max}], \tag{1}$$

where the support $[\beta_{\min}, \beta_{\max}]$ is the theoretically permissible range across submarkets given submarket primitives.

Although heterogeneity is the rule rather than the exception, empirical work rarely estimates individual β_i. Instead, researchers must aggregate over some submarkets and regress price changes on tax changes to identify an aggregate coefficient. To define this aggregate formally, let ω_i denote submarket i's pre-tax share of total transaction value, and define $\mathbb{E}_\omega[X] \equiv \int X(i)\, \omega_i \, dG(i)$ as the share-weighted average of a submarket-level quantity. The aggregate passthrough across the original pre-tax *cohort* of submarkets is:

$$\beta_{\text{cohort}} \equiv \mathbb{E}_\omega[\beta_i]. \tag{2}$$

The empirical complication is that not every submarket remains in the observed data following the tax change. At the new post-tax state, agents in each submarket compare the payoff from continuing in the observed tax base to the best available alternative. The alternative may take the form of switching to a less-taxed channel via evasion or avoidance, ceasing operation entirely, or any other action that removes the submarket from the observed data. Let $V^d(\xi_i)$ denote the post-tax payoff from continuing and $V^a(\xi_i)$ the maximum over all such alternatives. Submarket i stays if and only if $V^d(\xi_i) \geq V^a(\xi_i)$, captured by the survival indicator:

$$\sigma_i = \mathbf{1}\{V^d(\xi_i) \geq V^a(\xi_i)\}. \tag{3}$$

The exit share is defined as $\theta \equiv 1 - \mathbb{E}_\omega[\sigma_i]$.

Because exit removes submarkets from the data, a transaction-value-weighted regression mechanically identifies the share-weighted average among *survivors*, rather than the cohort:

$$\beta_{\text{survivor}} \equiv \frac{\mathbb{E}_\omega[\beta_i\, \sigma_i]}{\mathbb{E}_\omega[\sigma_i]}. \tag{4}$$

The cohort estimand averages over the full pre-tax population; the survivor estimand drops submarkets with $\sigma_i = 0$. This is problematic for *ex post* policy evaluation. The relevant welfare baseline is the original cohort, but the data deliver only survivors.[2]

Assumption 1 (Heterogeneity and Selective Exit)**.** *(i)* $\theta > 0$*; (ii)* $\text{Cov}_\omega(\beta_i, \sigma_i) \neq 0$.

Part (i) requires that some submarkets exit the observed tax base. Part (ii) requires that exit propensity covaries with passthrough. When either fails, $\beta_{\text{survivor}} = \beta_{\text{cohort}}$ and the framework reduces to standard incidence analysis. The empirical tariff literature illustrates both conditions. Researchers typically observe only direct trade between countries and cannot observe trade that continues indirectly via transshipment, in which exporters may reroute through third countries to avoid tariffs; this generates $\theta > 0$. Moreover, the exporters with the resources and incentives to transship may systematically be those with the market power to absorb tariffs rather than pass them through, generating a nonzero covariance between σ_i and β_i, something explored in an application of this framework later in the paper.

Proposition 1 (Selection wedge)**.** *The survivor and cohort passthrough estimands differ by the share-weighted covariance of submarket-level passthrough with survival:*

$$\beta_{survivor} - \beta_{cohort} = \frac{\text{Cov}_\omega(\beta_i, \sigma_i)}{\mathbb{E}_\omega[\sigma_i]} = \frac{\text{Cov}_\omega(\beta_i, \sigma_i)}{1-\theta}. \tag{5}$$

The wedge factors into a magnitude and a direction. The survival share $1-\theta$ scales the magnitude of the gap and is observable. The sign of $\text{Cov}_\omega(\beta_i, \sigma_i)$ determines the direction and is not. The asymmetry is fundamental: selection occurs on unobservables, so the primitive that governs β_i also governs σ_i, and neither their joint distribution nor the sign of their covariance is identified from the survivor sample. Recovering the direction therefore requires either a structural restriction or auxiliary empirical evidence, with the consequence that survivor-based estimates either overstate or understate cohort incidence depending on which side the covariance falls. The paper develops both kinds of evidence in an application to tariffs.

Remark. Exit need not operate through a single channel. For instance, a tax may cause some firms to shut down and others to exit through evasion. When non-survival partitions into K mutually

2. Both estimands hold weights fixed at the pre-tax distribution ω_i, isolating the passthrough channel. If weights are instead allowed to re-equilibrate post-tax, the share-weighted regression identifies an object that mixes passthrough with reallocation across surviving submarkets.

exclusive margins indexed $k = 1, \ldots, K$, with $\sigma_i^{(k)} = \mathbf{1}\{i \text{ exits via margin } k\}$ and $\sum_k \sigma_i^{(k)} = 1 - \sigma_i$, the wedge of Proposition 1 decomposes additively. Letting $\theta_k \equiv \mathbb{E}_\omega[\sigma_i^{(k)}]$ and $\bar{\beta}_k \equiv \mathbb{E}_\omega[\beta_i \sigma_i^{(k)}]/\theta_k$ denote the share and average passthrough of margin-k exiters,

$$\beta_{\text{survivor}} - \beta_{\text{cohort}} = \sum_{k=1}^{K} \theta_k \, (\beta_{\text{survivor}} - \bar{\beta}_k). \tag{6}$$

Each term inherits the sign of $\beta_{\text{survivor}} - \bar{\beta}_k$: margins that select low-passthrough exiters contribute positively, margins that select high-passthrough exiters negatively. The aggregate sign reflects which contribution dominates.

2.2 Bounds on Cohort Passthrough

The selection wedge identifies the source of the bias but does not by itself bound the cohort estimand. Combining it with the law of total expectation does, requiring only two empirical inputs—the exit share θ and the survivor passthrough β_{survivor}—alongside a single theoretical restriction on the sign of $\text{Cov}_\omega(\beta_i, \sigma_i)$. Letting $\bar{\beta}_{\text{exiter}} \equiv \mathbb{E}_\omega[\beta_i \mid \sigma_i = 0]$ denote the average passthrough among exiters, the cohort estimand decomposes as

$$\beta_{\text{cohort}} \;=\; (1-\theta)\,\beta_{\text{survivor}} \;+\; \theta\,\bar{\beta}_{\text{exiter}}, \tag{7}$$

so any bound on $\bar{\beta}_{\text{exiter}}$ delivers a bound on β_{cohort}. The remaining question is what bounds on $\bar{\beta}_{\text{exiter}}$ the structure of the problem permits.

Proposition 2 (Sign-restricted bound on cohort passthrough)**.** *Suppose Assumption 1 holds and $\beta_i \in [\beta_{\min}, \beta_{\max}]$ for all i. Then*

(i) If $\text{Cov}_\omega(\beta_i, \sigma_i) > 0$, then $\beta_{cohort} \in \big[(1-\theta)\,\beta_{survivor} + \theta\,\beta_{\min},\ \beta_{survivor}\big)$.

(ii) If $\text{Cov}_\omega(\beta_i, \sigma_i) < 0$, then $\beta_{cohort} \in \big(\beta_{survivor},\ (1-\theta)\,\beta_{survivor} + \theta\,\beta_{\max}\big]$.

The proof, in Appendix A.1.2, combines Proposition 1 with the law of total expectation.[3] The bound operationalizes the dual structure of the wedge. The exit share θ controls the width of each interval, with tight bounds when exit is small. The sign of $\text{Cov}_\omega(\beta_i, \sigma_i)$ selects which interval applies, placing the cohort below the survivor estimate in case (i) and above it in case (ii). Recovering β_{cohort} does not require knowledge of which specific submarkets exited, the structural

3. Without the sign restriction on $\text{Cov}_\omega(\beta_i, \sigma_i)$, the bound reduces to the worst-case interval in Manski (1989) for a bounded outcome under selective observation. The sign restriction partitions this interval at β_{survivor} and selects the relevant half. The structure parallels the monotone-treatment-selection assumption in Manski and Pepper (2003), applied to a survival-versus-cohort context rather than a treated-versus-untreated context.

parameters generating selection, or the joint distribution of submarket-level elasticities. The sign of the covariance and the aggregate exit share suffice.

One limitation of the proposition is that it requires a sign restriction on the global covariance. This restriction may be unavailable when offsetting margins push the wedge in opposite directions. Appendix A.1.3 states a disaggregated version (Corollary 3) that imposes sign restrictions margin-by-margin instead, partitioning exit margins into those selecting low-passthrough exiters and those selecting high-passthrough exiters. The resulting interval has endpoints scaled by margin-class shares θ^+ and θ^- rather than the aggregate θ, and is the version invoked later in the tariff application of Section 3.

A second limitation of Proposition 2 is that the bound is worst-case. One endpoint assumes the exiter pool is concentrated entirely at the extreme boundary of the passthrough distribution ($\beta_{\min}$ or $\beta_{\max}$), which is unlikely in practice. We can tighten the bound by imposing modest structure on how submarkets sort across the survival margin. Under threshold selection, the marginal exiter is the submarket exactly indifferent between staying and exiting, and its passthrough is identified directly from how the survivor average varies with the exit share. The result tightens the interval by replacing the survivor-average endpoint with a structural limit.

Corollary 1 (Sharpening under threshold selection)**.** *Suppose type ξ has a continuous distribution, σ_i is a monotonic threshold function of type, and $\beta(\cdot)$ is monotone and continuously differentiable. Let $\beta_{marginal}(\theta) \equiv \beta_{survivor}(\theta) - (1-\theta)\,\frac{\partial \beta_{survivor}(\theta)}{\partial \theta}$. Then*

(i) If $\mathrm{Cov}_\omega(\beta_i, \sigma_i) > 0$, then $\beta_{marginal} \le \beta_{survivor}$ and

$$\beta_{cohort} \in \left[(1-\theta)\,\beta_{survivor} + \theta\,\beta_{\min},\ (1-\theta)\,\beta_{survivor} + \theta\,\beta_{marginal}\right].$$

(ii) If $\mathrm{Cov}_\omega(\beta_i, \sigma_i) < 0$, then $\beta_{marginal} \ge \beta_{survivor}$ and

$$\beta_{cohort} \in \left[(1-\theta)\,\beta_{survivor} + \theta\,\beta_{marginal},\ (1-\theta)\,\beta_{survivor} + \theta\,\beta_{\max}\right].$$

Both intervals are strictly contained in the corresponding interval of Proposition 2 under Assumption 1.

The proof, in Appendix A.1.4, derives the tightened bounds via change of variables on the survivor integral and substitutes the result into the law of total expectation. As exit grows, the survivor average shifts because the marginal submarket leaves the sample. Differentiating the average with respect to the exit share isolates the passthrough at the boundary. Under monotone selection, the marginal exiter sets a tighter limit on the inframarginal exiter pool than the worst-case endpoint of Proposition 2. Since $\partial\beta_{\text{survivor}}/\partial\theta$ can be identified from cross-sectional variation in θ, β_{marginal} is recoverable from data alone, so no further structural input is needed beyond monotone threshold selection.[4]

2.3 Mapping Bias into Welfare

It is standard to evaluate the welfare cost of a tax instrument through the marginal value of public funds (MVPF) (Mayshar 1990; Bastani 2025), the first-order welfare cost per dollar of revenue raised. Endogenous exit enters the MVPF through both the numerator, which relates incidence to welfare, and the denominator, which captures how tax revenue responds to the tax rate. Appendix A.3 shows that the exit-based MVPF, which abstracts from non-exit margins, takes the form

$$\text{MVPF}_{\text{exit}} = \frac{\lambda_B\,\beta_{\text{cohort}} + \lambda_S\,(1-\beta_{\text{cohort}})}{(1-\theta)\,(1+\eta)}, \qquad \eta \equiv -\frac{\theta}{1-\theta}\,\psi, \qquad \psi \equiv \frac{\partial \ln \theta}{\partial \ln \tau}, \tag{8}$$

where τ is the tax rate at which revenue is collected, and λ_B and λ_S are the planner's welfare weights on buyers and sellers. For simplicity, I assume the planner applies uniform within-group weights; a richer specification would weight individuals separately, capturing heterogeneity in the social marginal utility of income.

The numerator is the planner-weighted welfare cost on the cohort, $B^{\text{cohort}}[\lambda_B\beta_{\text{cohort}} + \lambda_S(1-\beta_{\text{cohort}})]$. The passthrough is the cohort object because welfare is assessed over the population originally facing the tax, not just survivors. Empirical estimates deliver β_{survivor}, not β_{cohort}. The resulting error in the numerator is

$$B^{\text{cohort}}\,(\lambda_B-\lambda_S)(\beta_{\text{survivor}}-\beta_{\text{cohort}}),$$

the passthrough wedge scaled by the difference in weights and by the cohort base. The error vanishes when $\lambda_B = \lambda_S$, where the buyer and seller mismeasurements cancel, and grows as the weights diverge. The buyer term rises with β_{cohort} while the seller term falls, so without committing to which weight is larger, bounding the numerator requires bounding β_{cohort} from both sides. The bound-from-both-sides requirement disappears when one weight is zero. For example, a nationalist planner taxing foreign exporters places no weight on sellers, so the numerator collapses to $B^{\text{cohort}}\lambda_B\beta_{\text{cohort}}$ and the one-sided bound from Proposition 2 maps directly to it.

The denominator is the net revenue raised by a marginal increase in the tax, decomposed into a mechanical effect and a fiscal externality. Revenue collected on the surviving base is $R = \tau\,(1-$

4. Under multi-margin selection (equation (6)), each margin defines a boundary of the survivor region in type space; the empirical θ may correspond to one specific margin or to aggregate exit, and the recovered β_{marginal} should be interpreted as the marginal exiter at that margin.

$\theta)\, B^{\text{cohort}}$, where θ rises with τ as higher tariffs induce more exit. Differentiating, $\partial R/\partial\tau = (1-\theta)(1+\eta)\, B^{\text{cohort}}$: the mechanical term $(1-\theta)B^{\text{cohort}}$ collects the higher rate on the survivors, and the fiscal externality η nets against it as the base itself erodes. The common B^{cohort} factor cancels against the same term in the numerator, leaving the form in equation (8). The decomposition $\eta = -(\theta/(1-\theta))\psi$ isolates two pieces. The elasticity ψ measures how responsively the exit share moves with the tax. The share factor $\theta/(1-\theta)$, the ratio of exiters to survivors, scales that response, so the same elasticity counts for more when the exit margin is already large. With $\eta < 0$ whenever exit rises with the tax, the denominator collapses as $\eta \to -1$, a Laffer ceiling specific to the exit margin; non-exit Laffer mechanisms operate separately. The exit-based MVPF captures the welfare consequences of the exit margin alone; non-exit margins would add further terms to the fiscal externality in a fuller analysis but are outside the scope here.

The exit-based MVPF derived above is a sufficient-statistic correction in the tradition of Feldstein (1999) and Chetty (2009a), and it is worth elaborating on how mine compares to theirs. Feldstein established the taxable income elasticity as a sufficient statistic for the deadweight loss of the income tax. Chetty showed this fails when part of the marginal sheltering cost is a transfer rather than a real resource cost, so excess burden depends on the resource cost of sheltering. Both corrections rest on a correctly identified baseline elasticity that does not suffice for welfare, and use limited additional structure to recover the welfare-relevant object. They differ in where the insufficiency arises. In Chetty's, the elasticity estimates the right population but enters welfare weighted by the tax rate when the resource cost is the correct weight; here, β_{survivor} enters welfare correctly weighted but estimates the survivor average rather than the cohort average, and Proposition 2 bounds the latter. Chetty's resource-vs-transfer distinction does not enter the exit-based MVPF. The numerator is the first-order tax burden, with agents' optimal responses dropping out by envelope, leaving β for buyers and $1-\beta$ for sellers; the denominator is the marginal revenue response. The resource cost agents incur to exit appears in neither. It is a genuine welfare loss, but second-order and inframarginal. Chetty's object is built to price exactly this cost; the exit-based MVPF is not.

Within the exit channel, Propositions 1–2 and equation (8) together reduce welfare evaluation to a small set of empirical objects: the survivor passthrough β_{survivor}, the exit share θ, the exit elasticity ψ, and the sign of $\text{Cov}_\omega(\beta_i, \sigma_i)$. The first three are estimable from standard tax data. The fourth is not, and the framework alone is silent on it. Signing the covariance requires additional structure on the economic environment that generates exit. The next section develops this structure for an application to tariffs.

3 Tariff Passthrough under Endogenous Exit

The remainder of the paper applies the framework in Section 2 to an open puzzle: a large literature finds that the United States bore the full burden of the 2018 tariffs on China (Amiti, Redding, and Weinstein 2019, 2020; Fajgelbaum et al. 2020; Cavallo et al. 2021) despite standard theory predicting imperfect passthrough. This literature typically relies on customs data and therefore conditions on the survivor set of reported direct trade between China and the United States; it identifies β_{survivor} rather than the welfare-relevant β_{cohort}.[5] Whether β_{survivor} overstates or understates the cohort estimand depends on the sign of selection, which depends in turn on the joint structure of demand and exit. Under CES with homogeneous markups, β_i is constant across firms and $\beta_{\text{survivor}} = \beta_{\text{cohort}}$, so the survivor estimate is unbiased and the puzzle reflects the underlying parameter. Under Kimball, passthrough varies with marginal cost, and the survivor-cohort gap inherits the sign of selection on the cost distribution. Which direction selection runs depends on which exit margin dominates, and this section shows that exporters select on two opposing margins—rerouting and participation—that pull the cost distribution in opposite directions.

3.1 Environment

I work in a static partial equilibrium setting with monopolistic competition over differentiated varieties indexed by $i \in \mathcal{I}$. Each seller sets a price p_i and reaches U.S. buyers through one of two channels indexed by $r \in \{d, a\}$ (direct or rerouted), facing ad valorem tariffs $\tau^d \geq \tau^a \geq 0$. The gross-of-tax price under channel r is $\tilde{p}_i = (1 + \tau^r)p_i$.

Buyers. U.S. importers submit demand for each variety based on its gross-of-tariff price $\tilde{p}_i$ and a price aggregator $\mathbf{P}$ summarizing competing varieties within the product category,

$$q_i = D_i(\tilde{p}_i \mid \mathbf{P}).$$

$\mathbf{P}$ is exogenous at the firm level, corresponding to a standard monopolistic competition setup in which each firm is small relative to the market and takes the price distribution of rivals as given. Denoting the demand elasticity as $\varepsilon_i > 1$, I assume that demand is weakly Kimball (1995) with superelasticity

$$\kappa_i(\tilde{p}_i \mid \mathbf{P}) \equiv -\frac{\partial \ln \varepsilon_i(\tilde{p}_i \mid \mathbf{P})}{\partial \ln \tilde{p}_i} \leq 0,$$

5. Firm-level studies such as part of Cavallo et al. (2021) and Flaaen, Hortaçsu, and Tintelnot (2020) avoid origin-based selection by tracking specific goods.

which nests CES at $\kappa = 0$ and allows elasticity to rise with price when $\kappa < 0$. The Kimball specification is the standard non-CES generalization in the international pricing literature, motivated by evidence that markups and passthrough vary systematically with firm size and marginal cost in ways constant-markup CES demand cannot generate (Gopinath and Itskhoki 2010; Amiti, Itskhoki, and Konings 2019). The full set of regularity conditions on demand, **D1**–**D5**, is in Appendix A.2.1.

Sellers. Each foreign exporter $i \in \mathcal{I}$ has constant marginal cost m_i, drawn independently from a continuous distribution G with support $[\underline{m}, \bar{m}] \subset (0, \infty)$ and density $g(m) > 0$. The distribution is independent of policy and of the aggregator $\mathbf{P}$. Each exporter chooses among three options: operate the direct channel at tariff τ^d, pay an additional fixed cost $F_a > 0$ to reroute and operate at τ^a instead, or shut down. Operating either channel requires paying a fixed cost $F_d \geq 0$, standard in heterogeneous-firm trade models since Melitz (2003). The rerouting fixed cost F_a is sunk and captures the expenses of establishing the alternative channel: contracting with hub-country intermediaries, repackaging or relabeling goods, and bearing the legal and reputational risk of misdeclared origin. The rerouting tariff τ^a is the expected per-unit operating cost of the indirect path, combining hub-country tariffs, enforcement risk if the reroute is detected, and any quality or delay frictions. The baseline model treats F_a and τ^a as common across firms within an HS6 product; Appendix S.1.1 extends to product-specific F_a and Appendix S.1.2 to per-unit rerouting costs that depend on shipment size, and the qualitative results are preserved in both cases. Letting

$$V(m, 1+\tau^r) \equiv \max_{p \geq m} (p - m) D((1+\tau^r)p \mid \mathbf{P})$$

denote optimized variable profit in channel r, exporter i solves

$$\pi(m_i) = \max\left\{ V(m_i, 1+\tau^d) - F_d, \;\; V(m_i, 1+\tau^a) - F_a - F_d, \;\; 0 \right\}, \tag{9}$$

corresponding to direct, reroute, and shutdown. Define the avoidance gain $\Delta V(m) \equiv V(m, 1+\tau^a) - V(m, 1+\tau^d)$.[6]

6. I assign the rerouting decision to the exporter to preserve tractability. We could also see rerouting due to importers searching across origin countries or trade intermediaries arbitraging tariff differentials across different countries. In Appendix S.1.0.1, I show that under efficient bargaining, the cutoffs and the sign of $\mathrm{Cov}_\omega(\beta_i, \sigma_i)$ depend only on the joint surplus across channels, independent of which side bears F_a and F_d, so the tractability choice is unimportant under standard assumptions.

3.2 Pricing, Passthrough, and Selection

This subsection characterizes pricing within each channel, the resulting passthrough heterogeneity, and the partition of firms across the three options of Section 3.1.

Lemma 1 (Lerner condition, conditional on channel)**.** *For each channel $r \in \{d, a\}$ with corresponding wedge τ^r, under **D1–D4**, the optimal price $p_i^*(m_i; r)$ is interior and satisfies*

$$\frac{p_i^* - m_i}{p_i^*} = \frac{1}{\varepsilon_i(\tilde{p}_i^r \mid \mathbf{P})}, \qquad \tilde{p}_i^r = (1 + \tau^r)p_i^*.$$

Moreover: (i) p_i^* strictly increases in m_i; (ii) p_i^* is weakly decreasing in τ^r (constant under CES); and (iii) q_i^* strictly decreases in m_i and τ^r.

Proof. See Appendix A.2.2. □

The Lerner condition delivers the standard markup inversely proportional to the demand elasticity at the delivered price. Higher marginal costs imply higher prices, lower quantities, and lower markups within each channel. The key object for incidence is passthrough.

Proposition 3 (Passthrough heterogeneity, conditional on channel)**.** *Conditional on operating in the direct channel, holding* $\mathbf{P}$ *fixed, the passthrough of τ^d onto the gross price of variety i is*

$$\beta_i \equiv \frac{d \ln \tilde{p}_i}{d\tau^d} = \frac{1}{1 + \tau^d} \cdot \frac{\varepsilon(\tilde{p}_i) - 1}{\varepsilon(\tilde{p}_i) - 1 - \kappa(\tilde{p}_i)}, \tag{10}$$

where $\tilde{p}_i = (1+\tau^d)p_i^*(m_i)$. Under CES ($\kappa = 0$), passthrough is homogeneous across firms at $1/(1+\tau^d)$. Under Kimball ($\kappa < 0$), passthrough is strictly increasing in marginal cost, $\partial\beta_i/\partial m_i > 0$.

Proof. See Appendix A.2.3. □

The proposition decomposes passthrough into a mechanical factor $1/(1 + \tau^d)$ and a curvature factor $(\varepsilon - 1)/(\varepsilon - 1 - \kappa)$, with the latter strictly increasing in m_i under Kimball via the Lerner condition. The economic mechanism runs through demand curvature. Low-cost exporters charge deep markups at low delivered prices and sit in the relatively inelastic region of demand. A tariff increase pushes them toward the more elastic region and puts the most markup at risk, so they absorb most of it. High-cost exporters already operate in the more elastic region with thin markups and less to protect, and pass the tariff through.

With β_i heterogeneous in m_i, the partition of firms across the three options of Section 3.1 determines which firms supply the survivor sample and hence the gap between β_{survivor} and β_{cohort}. Establishing the partition requires that rerouting be expensive enough for exporters on the verge of shutting down not to finance it instead.

Assumption 2 (No re-entry)**.** *$F_a \geq \Delta V(\bar{m})$, where $\bar{m}$ is defined by $V(\bar{m}, 1+\tau^d) = F_d$.*

Without the assumption, very-high-cost firms would reroute rather than shut down, producing a non-monotone partition. The condition is vacuous when $F_d = 0$ or $\tau^a = \tau^d$. With the assumption in place, the partition is monotone in marginal cost: low-cost firms reroute, intermediate-cost firms ship directly, and high-cost firms shut down, as the next proposition formalizes.

Proposition 4 (Channel partition)**.** *Under **D1–D5** and Assumption 2, there exist unique cutoffs $\widehat{m}(\tau^d, \tau^a, F_a)$ and $\bar{m}(\tau^d, F_d)$ with $\widehat{m} \leq \bar{m}$ such that the cost distribution partitions almost surely as*

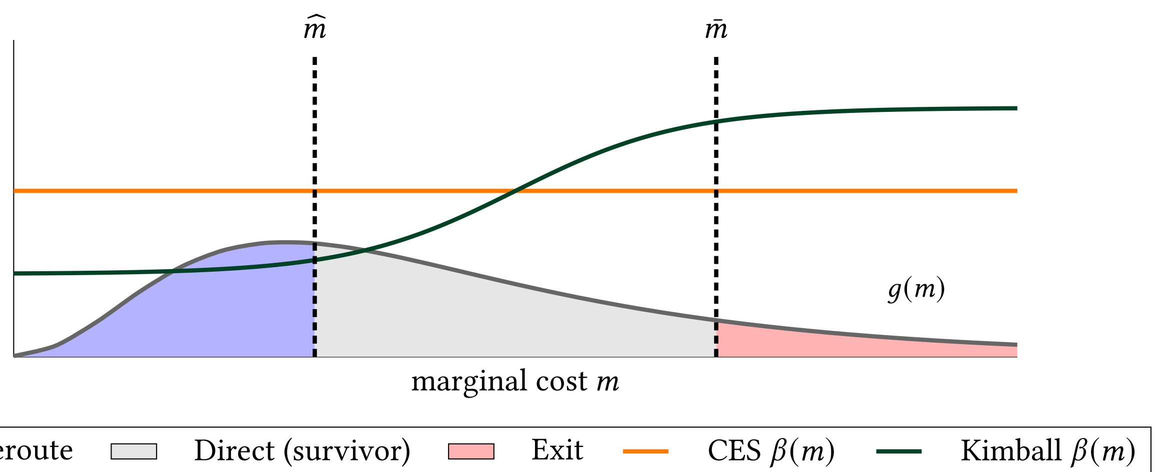
$$\begin{aligned} m < \widehat{m} &\Rightarrow \textit{Reroute}, \\ \widehat{m} \leq m \leq \bar{m} &\Rightarrow \textit{Direct (survivor)}, \\ m > \bar{m} &\Rightarrow \textit{Shut down.} \end{aligned}$$

At interior cutoffs, a tariff increase shrinks the survivor set from both sides:

$$\frac{\partial \widehat{m}}{\partial \tau^d} > 0, \qquad \frac{\partial \bar{m}}{\partial \tau^d} < 0.$$

The remaining cutoff sensitivities, $\partial \widehat{m}/\partial \tau^a < 0$ and $\partial \bar{m}/\partial F_d < 0$, are derived in Appendix A.2.4.

Figure 1: Channel Partition under Heterogeneous Passthrough



Note: The distribution $g(m)$ shows the density of marginal costs. Passthrough is shown under CES (flat line) and Kimball (upward-sloping line). Dashed lines show the cutoffs at the original tariff; solid lines show the cutoffs after a tariff increase.

Figure 1 visualizes the partition. The avoidance cutoff $\widehat{m}$ is set where the avoidance gain $\Delta V(m)$ equals F_a, and the participation cutoff $\bar{m}$ where direct-channel variable profit $V(m, 1+\tau^d)$

equals F_d.[7] The two cutoffs are driven by different forces. Low-cost exporters absorb tariffs and lose substantial surplus when tariffs rise, which they recover by paying F_a to reroute at the lower wedge τ^a. High-cost exporters pass tariffs through and lose less per shipment, but their profit base is thin, so the same tariff increase pushes the marginal participant below F_d and they shut down.

The participation margin is the standard Melitz-style exit mechanism. In heterogeneous-firm trade models with fixed export costs, the least productive firms exit when costs rise. The avoidance margin, typically absent from those models, runs in the opposite direction. Formally, the two-margin partition specializes equation (6) of Section 2 to $K = 2$, with avoidance ($k = a$) selecting low-passthrough exiters and participation ($k = p$) selecting high-passthrough exiters, and the framework's prediction depends on which margin dominates.

Corollary 2 (Sign of the survivor-cohort gap)**.** *Let $\theta_a = \Pr(m < \widehat{m})$ and $\theta_p = \Pr(m > \bar{m})$, and let $\bar{\beta}_a, \bar{\beta}_p$ denote the corresponding within-region average passthroughs. Then $\beta_{survivor} > \beta_{cohort}$ if and only if*

$$\theta_a(\beta_{survivor} - \bar{\beta}_a) > \theta_p(\bar{\beta}_p - \beta_{survivor}). \tag{11}$$

Proof. See Appendix A.2.5. □

The partition also pins down the structural content of the exit elasticity ψ entering the MVPF denominator of Section 2.3. With two margins operating, ψ aggregates margin-specific elasticities ψ_a and ψ_p share-weighted by their contributions θ_a/θ and θ_p/θ; Appendix A.3 formalizes the aggregation. Both elasticities are strictly positive by Proposition 4. Holding τ^a fixed at its policy-invariant level, an increase in τ^d raises both θ_a (by widening the avoidance gap $\Delta = \tau^d - \tau^a$) and θ_p (by pushing the marginal participant below F_d). Under avoidance dominance $\psi \approx (\theta_a/\theta)\,\psi_a$, and the welfare evaluation collapses to one structural object, namely the elasticity of the rerouting share with respect to Δ. Greater demand curvature widens the passthrough gap between low- and high-cost exporters, amplifying the profit differential at each cutoff and raising the responsiveness of both margins. More curvature, more selection.

What remains is empirical. Section 4 develops a methodology to measure θ_a and θ_p in customs data; Section 5 reports the magnitudes and verifies that avoidance dominates.

4 Measuring Exit in Customs Data

The 2018 tariffs levied by the United States on China gave Chinese exporters an incentive to exit the observed customs tax base, whether by rerouting trade through third countries or by

7. Corollary 1 of Section 2 assumes one-sided threshold selection and so applies to each margin separately, with $\widehat{m}$ as a lower threshold and $\bar{m}$ as an upper threshold. The disaggregated bound of Corollary 3 combines the two.

shutting down entirely. Section 3 showed that the gap between survivor and cohort passthrough depends on this selection, but its magnitude is an empirical question. How much trade left the direct China–U.S. channel after 2018, and through which margin? This section measures both margins in customs data, which record the tax base as the taxing authority itself observes it. The two leave different footprints. Rerouted goods reappear under a third-country declared origin, so transshipment must be recovered by triangulating China–hub–U.S. flows. Shutdown leaves a one-sided decline in direct trade with no offsetting hub flow. I first make explicit how the firm-level model maps to product-level customs data, then construct the two margins.

4.1 From Firms to Products

The model develops firm-level objects; the data are at the product level. The mapping is as follows.

As in Section 3, each firm takes the product-level price aggregator $\mathbf{P}$ as given, so passthrough is a function of marginal cost alone and exit shifts which firms survive without repricing those that remain; the survivor–cohort gap reflects selection rather than the strategic response to thinner competition that oligopolistic passthrough would generate. Each product k consists of a continuum of firms drawing marginal costs from a continuous distribution G_k with density $g_k > 0$. Product k faces tariffs τ_k^d in the direct channel and τ_k^a in the rerouting channel, with fixed costs F_a and F_d. By Proposition 4, firms partition into three regions at the avoidance cutoff $\widehat{m}_k \equiv \widehat{m}(\tau_k^d, \tau_k^a, F_a)$ and the participation cutoff $\bar{m}_k \equiv \bar{m}(\tau_k^d, F_d)$. The product-level avoidance share is $\theta_{a,k} \equiv G_k(\widehat{m}_k)$, the participation share is $\theta_{p,k} \equiv 1 - G_k(\bar{m}_k)$, and the survivor passthrough is $\beta_{\text{survivor},k} \equiv \mathbb{E}_{G_k}[\beta(m) \mid \widehat{m}_k \leq m \leq \bar{m}_k]$.

The framework from Section 2 applies within each product, with $\theta_k = \theta_{a,k} + \theta_{p,k}$ as the within-product exit share. The sign of the within-product covariance, and hence the direction of the bound, is determined empirically in Section 5.2. Aggregating across products with 2017 import shares as weights ω_k yields share-weighted aggregates $\beta_{\text{cohort}} \equiv \mathbb{E}_\omega[\beta_{\text{cohort},k}]$ and $\beta_{\text{survivor}} \equiv \mathbb{E}_\omega[\beta_{\text{survivor},k}]$.

Three empirical inputs are needed at the product level: $\theta_{a,k}$, $\theta_{p,k}$, and $\beta_{\text{survivor},k}$. The first two are constructed from the measurement methodologies developed below; the third is taken from customs-based price regressions in the literature (Amiti, Redding, and Weinstein 2019; Fajgelbaum et al. 2020; Cavallo et al. 2021). The firm-level distribution G_k and the cutoffs $(\widehat{m}_k, \bar{m}_k)$ are not separately identified, and within-product heterogeneity in $\beta(m)$ is unobserved. A product-level regression identifies the structural $\beta_{\text{survivor},k}$ only if the within-cell variety mix is stable; Section 5.2 examines this channel directly.

4.2 Data

I use annual data on bilateral trade flows from CEPII's BACI HS-12 vintage from 2012–2023 at the 6-digit product (HS6) level.[8] For each year t, country pair (i, j), and HS6 code k, I observe the customs value of shipments $v_{i \to j,k,t}$. The goal is to identify when goods originate in China, flow to a hub country on the first leg, and then arrive in the United States on the second leg. I restrict potential hubs to 36 countries selected for geographic proximity to China, port infrastructure, and pre-tariff trade relationships with both the U.S. and China (Table B.1). I deflate all values to 2017 dollars using the BLS end-use import price index.

The tariff wedge Δ operationalizes the theory's routing incentive as the gap between the effective cost of direct shipping and rerouting. U.S. importers deduct tariff-inclusive costs from taxable income at the corporate rate τ^c_t. Consumption and intermediate goods are expensed immediately, so the full cost is deducted in the year of import; capital goods are depreciated over time. Let $z_{k,t}$ denote the present value of deductions per dollar of cost, with $z = 1$ for expensed goods and $z < 1$ for capital. Paying the direct tariff thus generates a deduction worth $z_{k,t}\tau^c_t$ per dollar of tariff. The cost a firm forgoes by rerouting is the effective rerouting rate,

$$\tau^a_{k,t} = z_{k,t}\, \tau^c_t\, \tau^d_{k,t}.$$

This produces heterogeneity in τ^a across products, something developed theoretically in Appendix S.1.3. The wedge is

$$\Delta_{k,t} \equiv \log \frac{1 + \tau^d_{k,t}}{1 + \tau^a_{k,t}} = \log \frac{1 + \tau^d_{k,t}}{1 + z_{k,t}\tau^c_t \tau^d_{k,t}}.$$

For the direct tariff, I use Amiti, Redding, and Weinstein (2020) for monthly HS10 China-specific rates through 2017, aggregated to annual HS6 using 2017 import shares as weights.[9] From 2018–2023, I use the Global Tariff Database from Rodríguez-Clare, Ulate, and Vasquez (2025), which provides bilateral HS6 tariffs based on the methodology in Teti (2025).

I classify each HS6 product as consumption, intermediate, or capital using the UN's Broad Economic Categories (BEC). For consumption and intermediate goods, $z_k = 1$. For capital goods, $z_{k,t}$ is the net present value of depreciation deductions under MACRS.[10] I map each capital good

8. BACI provides consistent bilateral flows at annual frequency with comprehensive country coverage. Monthly UN Comtrade data have gaps, reporting lags, and reconciliation issues across partner countries that would complicate the screening methodology. The HS6 level is the finest internationally harmonized classification in bilateral trade data; finer classifications are country-specific.

9. Trade-weighted aggregation reflects the tariff rate on goods actually traded. Results are robust to alternative aggregation schemes including time-weighting and simple averages.

10. For an asset with tax life T years and depreciation schedule $\{d_s\}_{s=0}^{T}$, the NPV of deductions at discount rate

to its IRS tax life and compute $z_{k,t}$ accordingly. Table C.1 shows the pre- and post-2018 means of the resulting wedge $\Delta_{k,t}$ by end-use category. Variation across end-use categories is driven primarily by differences in statutory tariff rates τ^d rather than by differences in deductibility z.

Products that span multiple BEC categories are included once per category, with regression weights equal to the product's BEC category share multiplied by its share of 2017 China-U.S. imports. This ensures each product contributes to estimates for all relevant end-use categories in proportion to its actual composition.

4.3 Constructing the Exit Share

The exit share $\theta = \theta_a + \theta_p$ has two components, each measured separately. The avoidance margin shows up as transshipment, trade rerouted through hubs, which I recover by screening bilateral flows. The participation margin shows up as shutdown, the cessation of direct trade, which I recover from sustained absence net of rerouting. I construct each in turn.

4.3.1 The Transshipped Share

Measuring transshipment is inherently difficult. Unlike domestic tax evasion, it leaves no audit trail or third-party report. I construct bounds on θ_a by screening bilateral trade data for patterns consistent with Chinese goods flowing through third countries en route to the United States. The approach extends Freund (2024), who develops conditions to identify transshipment in bilateral trade data; I tighten these conditions and calibrate thresholds to control the false positive rate. The methodology complements firm-level evidence from Vietnamese customs records (Iyoha et al. 2025) and bill-of-lading data (Do et al. 2025).

The methodology imposes five conditions that a product-hub-year triplet must satisfy simultaneously. First, the product must have faced U.S. tariff increases on China, which restricts attention to goods with a policy-induced incentive to reroute. Second, China's share of U.S. imports must have fallen below its pre-trend while a hub country's share rose above its own, a reallocation pattern consistent with rerouting. Third, China's exports to the rest of the world must have remained stable relative to pre-trends, ruling out the alternative that China lost global competitiveness in the product. Fourth, the hub must face a lower U.S. tariff than China on that specific product, ensuring an economic incentive to reroute through that hub. Fifth, China-to-hub trade flows must be large enough to plausibly supply the hub-to-U.S. surge. If Vietnamese firms were expanding domestic production rather than re-exporting Chinese goods, Chinese shipments to Vietnam would show no corresponding increase for the same products in the same years.

r is $z = \sum_{s=0}^{T}(1+r)^{-s}d_s$. With bonus depreciation share b, this becomes $z_{\text{bonus}} = b + (1-b)z$. Under 100% bonus depreciation (post-TCJA), $z = 1$.

The screens are stringent by design. Each condition is evaluated against product-specific pre-trends estimated from 2012–2017, so a triplet is flagged only when trade patterns deviate sharply from historical norms. I construct two screens. The conservative screen flags only above-trend growth on both legs of the route; the liberal screen also captures scaling of pre-existing routes. Both are calibrated against a placebo sample of products that never faced tariff increases. I select thresholds such that the implied transshipment share among these never-treated products does not exceed 1% in any pre-tariff year. This calibration prioritizes specificity over sensitivity, minimizing false positives at the cost of missing some genuine rerouting.

The measure θ_a captures trade that left the direct China-U.S. channel and reappears in matched hub flows. Pure participation exit, where a Chinese firm stops exporting with no offsetting activity, produces a one-sided decline in direct trade and is not flagged by the screens, which require both a drop in China's U.S. share and a coincident rise in a hub's U.S. share, backed by growth in the China-to-hub leg.

The placebo calibration above bounds the overall false-positive rate at under one percent. The measure is a lower bound on total avoidance, since it omits margins such as value misreporting and product misclassification. Implementation details are in Appendix B.

4.3.2 The Shutdown Share

The participation margin removes a Chinese exporter from the direct channel entirely. In product-level data this shows up as direct China–U.S. trade that vanishes without reappearing through a hub. I identify shutdown positively, as products whose direct trade vanished and stayed vanished, rather than as a residual. A residual would absorb within-survivor declines and substitution by U.S. buyers toward non-Chinese suppliers; a positive criterion can be built to exclude both.

A code is flagged as shutdown if three conditions hold. First, the ratio of its direct China–U.S. trade to its 2012–2017 pre-trend forecast, formed as in the transshipment screen, falls below a threshold $\underline{\epsilon}$ in 2018, 2019, and 2023, so that direct trade vanished early in the tariff period and remained absent through the reporting year rather than dipping transitorily. Second, China's exports of the product to the rest of the world fell below their own pre-trend over the same years. Genuine firm exit lowers China's global shipments of the product; a vanished U.S. flow alongside a healthy global one points instead to U.S. buyers substituting toward other origins, which is not exit. Third, the transshipment screens flag no rerouting for the code in any post-period year, removing trade that left the direct channel but reappeared through a hub. The threshold $\underline{\epsilon}$ is calibrated against the same placebo sample of never-treated products as the transshipment screen, holding the implied shutdown share among those products below one percent in any pre-tariff year.

The three conditions make shutdown and transshipment disjoint by construction, so a code enters at most one of θ_a and θ_p. Because the carve-out conditions on the transshipment screen, the shutdown share inherits the conservative–liberal distinction; the two versions differ only in which already-rerouted codes are excluded, and they yield nearly identical shutdown shares. The measure is an upper bound on the participation margin. The conditions cannot separate firm exit from production relocated out of China, since relocation lowers China's exports to the United States and to the rest of the world alike; bilateral trade data record where goods ship from, not whether the firm behind them has closed. They do exclude within-survivor decline, which leaves direct trade positive, and U.S.-side substitution, which leaves China's global exports intact.

5 Measured Exit in Customs Data

With the two exit margins constructed, I now supply the two empirical inputs the bound of Proposition 2 requires: the magnitude of exit, which scales the correction, and the sign of $\mathrm{Cov}_{\omega}(\beta_i, \sigma_i)$, which selects its direction. Those, together with an estimate for the tariff elasticity of exit, are sufficient to also bound exit-based welfare, which is what I do in the following section.

5.1 The Magnitude of Exit

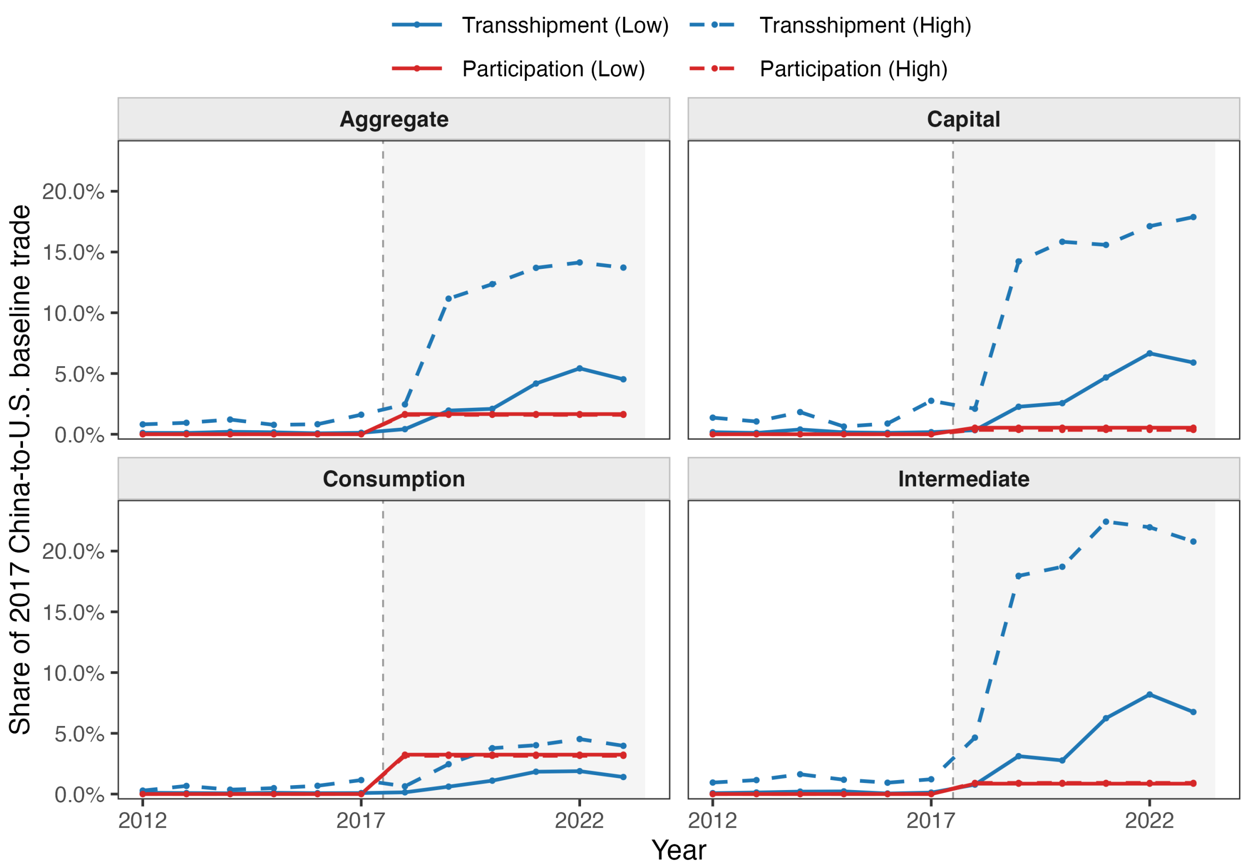
The first step to bounding incidence is measuring how large exit was in the wake of the 2018 trade war. This subsection measures that first using time series data, then by leveraging cross-sectional variation in exposure to tariffs across products to estimate the semi-elasticity of exit.

5.1.1 Exit in Levels

Figure 2 traces the magnitude of both exit margins—transshipment and shutdown—from 2012 through 2023, each as a share of 2017 baseline trade with China. Both sit near zero before 2018, with no pre-existing trend. Transshipment then breaks upward with the tariffs and climbs; shutdown steps up once and stays flat. Aggregate transshipment climbs to between 5 and 13 percent of the original tax base by 2023, depending on the screen, while shutdown levels off below two percent.

The aggregate averages over products that faced very different tariff increases, and the two margins fall differently across them. Transshipment concentrates where the wedge is largest: under the liberal screen it reaches over 20 percent of baseline trade for intermediates and 18 percent for capital goods by 2023, against less than 5 percent for consumption goods, in part because consumption goods saw far milder tariff increases. However, studies which track consumption goods directly from exporter to importer, such as Flaaen, Hortaçsu, and Tintelnot (2020) and parts of Cavallo et al. (2021), similarly find complete passthrough. Shutdown is small in all end-uses and concentrated in consumption goods, with a rise in magnitude paralleled by transshipment.

Figure 2: The Two Margins of Exit by End Use



Notes: Transshipment (θ_a) and shutdown (θ_p) as a share of 2017 baseline direct China→US trade, value-weighted, among treated HS6 codes (those facing U.S. tariff increases on China). Panels show the aggregate and the three end-use categories. Solid lines are the high screen, dashed the low. See Section 4.3 for more detail.

θ, which supplies the correction to incidence, rests almost entirely on transshipment rather than shutdown. Shutdown never exceeds two percent of baseline trade in the aggregate, and is an order of magnitude smaller than transshipment across all end-uses except consumption goods. Importantly, the magnitude of transshipment is also corroborated by outside evidence. For example, Iyoha et al. (2025) use firm-level Vietnamese customs data and find that rerouting accounts for 8.8 percent of Vietnam's growth in U.S.-destined exports over 2018–2021, and my two screens bracket that figure, implying 5.1 percent under the conservative screen and 14.2 percent under the liberal one over the same window. At 5 to 13 percent of the tax base, exit is large enough to move measured passthrough materially, and it runs almost entirely through transshipment, which the rest of the section takes up.

5.1.2 The Semi-Elasticity of Transshipment

The time series in Figure 2 show that transshipment rises when the tariffs hit. However, a time series alone cannot establish that tariffs were the cause. In this subsection, I use cross-sectional variation in exposure to tariffs to show that tariffs does cause transshipment to rise. In particular, I estimate the semi-elasticity of transshipment with respect to tariffs by running

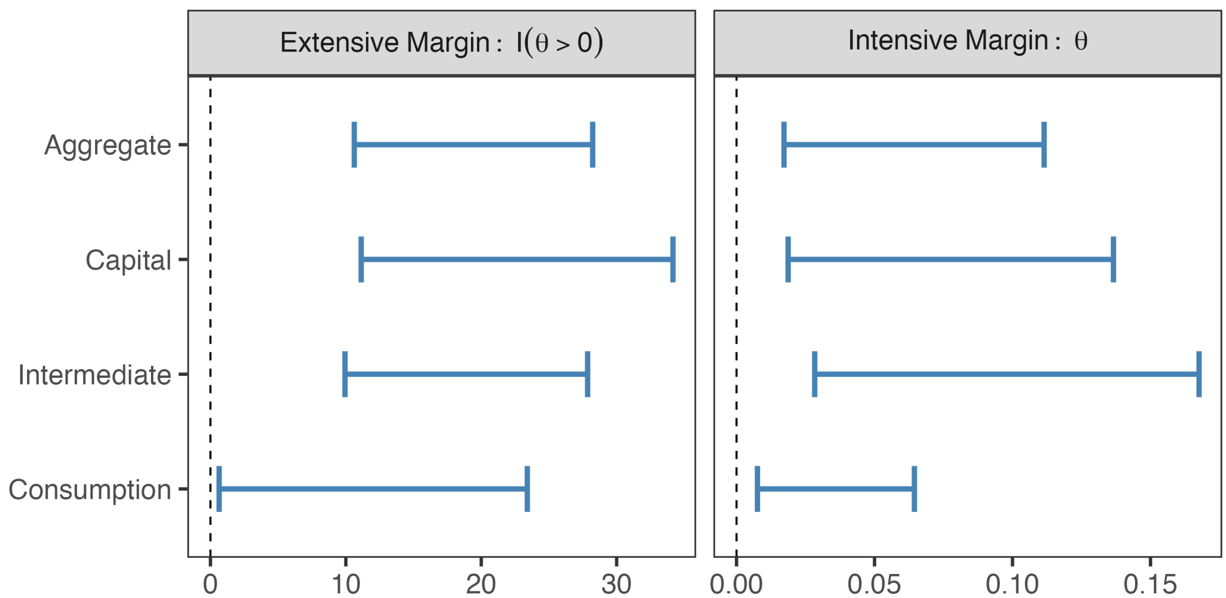
$$\theta_{k,t} = \alpha_k + \delta_{t\times e(k)} + \gamma \cdot \Delta_{k,t} + \varepsilon_{k,t}, \tag{12}$$

where the outcome is $\theta_{k,t}$ on the intensive margin or $\mathbf{1}(\theta_{k,t} > 0)$ on the extensive margin, α_k are HS6 fixed effects, $\delta_{t\times e(k)}$ are year×end-use fixed effects, observations are weighted by 2017 China→US import shares, and standard errors are clustered by HS6. The coefficient $\gamma \equiv \partial\theta/\partial\Delta$ is the semi-elasticity of transshipment with respect to tariffs. The two sets of fixed effects fix what variation identifies γ. The HS6 effects absorb everything time-invariant about a product, and the year×end-use effects absorb anything common to an end-use category in a given year, including the broad path of the tariff war within that category. What remains is within-product, within-end-use-year variation in the wedge, and γ is causal under the assumption that the products receiving larger wedge increases were not already trending toward more rerouting for reasons unrelated to the tariffs. I report Imbens and Manski (2004) joint 95% confidence sets throughout, taking account of both the liberal and conservative variants of the transshipment measure.

Figure 3 plots Imbens-Manski bounds at the aggregate level and by end-use. On the intensive margin, a 10 percentage-point increase in the wedge raises the rerouted share by 2 to 11 percentage points in aggregate; on the extensive margin, it raises the probability that a product reroutes at all by 11 to 28 percentage points. The aggregate masks a clear ordering across end-use which follows the same ordering as the levels in Figure 2. Intermediates respond most, with intensive-margin bounds of 3 to 17 percentage points for a ten percentage point increase in tariffs, against 2 to 14 for capital and under 1 to 6.5 for consumption. Business inputs reroute three to four times as readily as consumer goods. Appendix Table C.2 reports results on both the intensive and extensive margins separately for each measure of transshipment.

Several falsification tests bear directly on the identifying assumption. First, I conduct an event study in Appendix C.1, showing that transshipment only rises and remains elevated when the trade war begins. I reinforce this with a block permutation test, which reassigns treated status across products within HS4 groups and finds the pre-period gap in transshipment between treated and placebo codes no larger than chance (Appendix Table C.3). The treated products were not already drifting apart from the controls before the tariffs, so the divergence after 2018 is not pre-existing difference surfacing on its own. Second, I check whether the estimated relationship could arise from chance assignment of wedges by reassigning each product's full wedge trajectory to a random donor 5,000 times and re-running the main specification. Appendix Figure C.2 shows that the resulting coefficients center on zero and do not overlap with the true coefficients for either screen on either margin. Finally, I conduct a placebo test by applying the exact same screening criteria to the European Union and Canada, neither of which was directly involved in the trade war. Appendix Table C.4 shows that the resulting coefficients are generally indistinguishable from zero. This indicates that transshipment responds to the wedge only where the tariffs applied, and is not an artifact of a global shift in China's exports.

Figure 3: Exit Response to the Tariff Wedge



Notes: Imbens–Manski joint 95% confidence sets for γ in equation (12), aggregate and by end-use. Left panel: extensive margin (effect on the probability of any rerouting). Right panel: intensive margin (effect on the rerouted share θ).

The Supplemental Appendix contains several tests showing that the main coefficients are robust to alternative specifications. First, entropy balancing against pre-period imbalance (Table S10) and HS4 linear trends (Table S8) yield coefficients close to the baseline, as do slope-adjusted and long-differenced specifications (Tables S6 and S7). Additionally, alternative clustering at HS4×year (Table S1) and using more comprehensive fixed effects (Table S4), as well as unweighted estimation (Table S2), and a 2016–2019 window (Table S9) produce similar estimates. Importantly, leave-one-out exclusion of individual hubs (Table S11) shows the largest sensitivity to Vietnam on the extensive margin (45% change) and to Mexico on the intensive margin (33% change), consistent with rerouting concentrating in specific corridors. However, no exclusion flips the sign or significance of the main coefficients.

5.2 Signing the Covariance Between Exit and Passthrough

The sign of the covariance between passthrough and exit determines whether survivor-based estimates overstate or understate cohort incidence. Two exit margins pull it in opposite directions. Transshipment removes the low-cost, low-passthrough firms, the ones who absorb tariffs and gain most from the lower wedge, which would leave survivors selected toward high passthrough and make the covariance positive; participation exit removes high-cost, high-passthrough firms and would pull it negative. Section 5.1 found participation negligible, so by Corollary 2 the aggregate sign inherits the transshipment margin alone, and the model predicts it is positive. This section establishes whether or not that holds in the data.

The first test compares the passthrough of surviving firms across products that rerouted to different degrees. If the firms that reroute are the low-passthrough ones, products where more firms left should show higher passthrough among those who stayed. Such a gradient can only appear where passthrough varies across firms to begin with, which under the model is differentiated goods; homogeneous-markup goods have no passthrough heterogeneity for selection to act on and serve as a control. To run the test, I sort products into quartiles of post-2018 exit intensity and estimate survivor passthrough within each. The results are in Table 1. Within differentiated goods, survivor passthrough rises with exit intensity, while in homogeneous goods it shows no gradient. That the firms who stayed pass through more precisely where more firms left is what it means for the exiters to have been the low-passthrough firms, which is the positive covariance the model predicts and the opposite of Melitz. The same conclusion holds under cruder proxies for product type. For example, the wedge-exit response is two to three times larger where markups are most dispersed (Appendix Table C.12), and products in which China held a larger pre-tariff share reroute more (Appendix Table C.9). Neither pins the direction the way the gradient does, but both show exit selecting on type rather than firing at random.

The gradient establishes which products leave but not why. The model says there is transshipment because shipping indirectly yields higher profits, and that mechanism leaves its own footprint. Where rerouting through a hub is cheap, the wedge should drive a large exit response; where the hub itself faces a high U.S. tariff, the option is worth little and the response should fade. To test this, I add the interaction $\Delta_{k,t} \times \tau_k^h$ to equation (12), where τ_k^h is the U.S. tariff the candidate hub faces, and plot the implied response of exit to the wedge, $\partial\theta/\partial\Delta = \gamma + \rho\, \tau_k^h$, across the observed range of hub tariffs in Figure 4.

Table 1: Tariff Passthrough by Exit-Intensity Quartile

	$\ln P^{\text{imp}}_{k,t}$		
	Pooled	Differentiated	Non-Differentiated
	(1)	(2)	(3)
$\Delta_{k,t}$ (Q1 reference)	-0.207	-0.842	0.659
	(0.577)	(0.832)	(0.662)
$\Delta_{k,t} \times \mathbf{1}\{\bar{\theta}_k \in Q2\}$	1.48**	2.48***	-0.851
	(0.608)	(0.851)	(0.593)
$\Delta_{k,t} \times \mathbf{1}\{\bar{\theta}_k \in Q3\}$	1.78***	2.72***	0.500
	(0.596)	(0.841)	(0.525)
$\Delta_{k,t} \times \mathbf{1}\{\bar{\theta}_k \in Q4\}$	1.55*	1.77**	2.26
	(0.928)	(0.878)	(1.90)
R^2	0.95	0.95	0.97
Observations	50,245	33,505	16,740
HS6 & Year×Use FE	✓	✓	✓
Weighted	✓	✓	✓

* p < 0.1, ** p < 0.05, *** p < 0.01

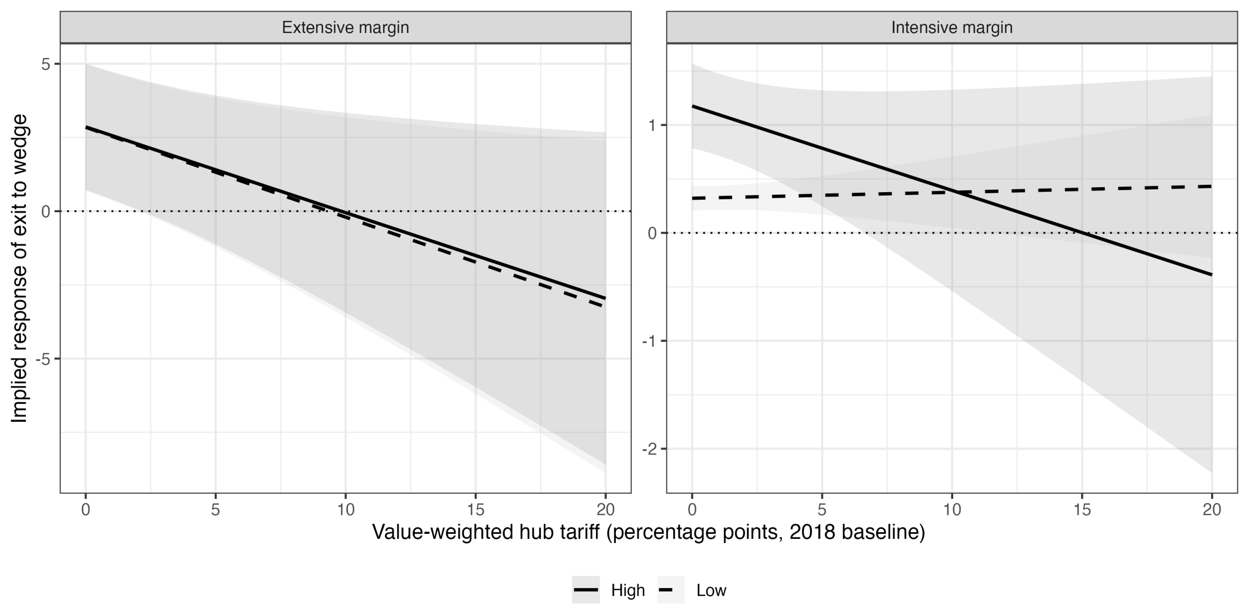
Notes: Outcome is the importer-paid log price $\ln P^{\text{imp}}_{k,t} = \ln P^{\text{CN}}_{k,t} + \Delta_{k,t}$. $\bar{\theta}_k$ is the post-2018 mean of $\theta^{\text{high}}_{k,t}$, partitioned into quartiles. Column (1) pools all treated HS6 codes; column (2) restricts to differentiated goods (Rauch "n"); column (3) is the non-differentiated commodity placebo. Sample is HS6 codes with positive 2019 wedge; observations with year-over-year $|\Delta \ln p_{\text{CN}}|$ outside [1%, 99%] are dropped. All specifications include HS6 and year×end-use fixed effects, weight by 2017 China→US exposure × pre-period end-use share, and cluster standard errors by HS6.

The prediction holds, most cleanly on the extensive margin. The implied response of exit to the wedge is 2.9 at a zero hub tariff under both screens and declines steadily as the hub tariff rises, crossing zero by ten percentage points. The slope against the hub tariff is the same whichever screen defines exit, so the decline is not an artifact of the measure. The intensive margin tells the same story under the liberal screen, falling from 1.2 at a zero hub tariff to zero near fifteen points, but is flat under the conservative screen, so the intensive evidence rests on one screen alone.[11]

11. Even so, the intensive-margin response varies with pre-2017 proxies for the fixed cost of rerouting in the

The decision to reroute at all is the one the cost margin should govern, and on that margin the response moves with rerouting cost under both screens. A story in which low-passthrough firms simply exit more, for reasons unrelated to the indirect channel, gives no reason for the response to fade as rerouting grows expensive.

Figure 4: Exit Elasticity by Rerouting Friction



Note: Implied exit elasticity to the tariff wedge, plotted against the candidate-hub US tariff on each HS6 code (τ_k^h, held fixed at 2018 values). Panels show conservative (Low) and liberal (High) screens on the intensive and extensive margins. Shaded bands are 95% pointwise confidence intervals from the delta method; the dashed line marks $\psi = 0$. Underlying specification extends equation (12) with $\Delta_{k,t} \times \tau_k^h$.

One measurement story could produce the same gradient in survivor passthrough with no firm-level selection behind it. Because β_{survivor} is estimated from an HS6-level price regression, it tracks the cell average rather than any individual variety, and a comparison before and after rerouting is a comparison of two different bundles of varieties. The coefficient is a pre-tax-weighted average, and the regression recovers a clean measure of survivor passthrough only if the variety mix within a cell holds steady. If high-value varieties drop out mechanically when a product reroutes, the cell average moves for compositional reasons that have nothing to do with the passthrough of the firms that remain. Without variety-level data I cannot rule the channel out or quantify what it leaves behind, but I can test its directional prediction. Mechanical exit of high-value varieties would thin the upper tail of the within-cell unit-value distribution. The data

predicted directions (Appendix Table C.5): products with greater pre-existing hub exposure and electronics content respond more, bulky goods less. The extensive-margin analogues share these signs but are imprecise (Appendix Table C.6).

move the other way. Appendix Table C.11 shows that higher rerouting is associated with a 15.7 percentage-point fall in the tenth percentile of unit value and a 7.6-point fall in the twenty-fifth, while the seventy-fifth and ninetieth show no significant shift. The distribution compresses from below, which is the wrong tail for pure composition. Appendix Table S12 shows that within-cell dispersion does not modulate the price-θ relationship either, so composition cannot be doing the work.

The exiters are the low-passthrough firms, they leave through the rerouting channel the model identifies, and the survivor passthrough they leave behind is selected rather than composed. The covariance is therefore positive, so the case-(i) bound of Proposition 2 applies. The next section carries that bound into the numerator of the marginal value of public funds and prices the welfare cost of the tariffs.

6 Corrected Tariff Incidence and the MVPF

In Section 2 I showed that correcting incidence relies on two inputs, the level of exit θ and the covariance between exit and passthrough. That corrected incidence is itself the numerator of the marginal value of public funds, and the same exit enters the denominator through a third input, the tariff elasticity of exit. With all three now supplied by the previous section, I bound measured tariff incidence and then the marginal value of public funds.

6.1 Corrected Incidence

With the level of exit and its direction in hand, I correct the survivor passthrough estimate for the firms that left. Since the covariance is positive, cohort passthrough lies below survivor passthrough, and case (i) of Proposition 2 gives the bound. Since participation exit is negligible, I apply the disaggregated form of the bound with the participation margin set to its framework-consistent upper bound, where participation exiters are high-cost firms passing through fully, so the correction runs through avoidance alone.

To illustrate the correction, I anchor survivor passthrough at $\beta_{\text{survivor}} \approx 1$, the value estimated across a wide range of customs-based studies (Amiti, Redding, and Weinstein 2019; Fajgelbaum et al. 2020; Amiti, Redding, and Weinstein 2020; Cavallo et al. 2021). I then vary the one object the data leave open, the passthrough β_{exiter} the exiting submarkets would have shown had they remained, from a floor of zero up to $\beta_{\text{exiter}} = 1$, where exiters look just like survivors and the correction vanishes. The positive covariance signed in Section 5.2 places the true value in the lower part of this range, since exiters pass through less than survivors.[12]

12. Ideally I would use Corollary 1 to tighten the floor, replacing the zero-passthrough exiter with the marginal one

Table 2 reports implied cohort passthrough as β_{exiter} ranges over its support. Each column fixes a value for the passthrough the exiting submarkets would have shown had they remained, and reads off the cohort average that value implies through equation (7). The floor at $\beta_{\text{exiter}} = 0$ holds exiters to zero passthrough and gives aggregate cohort passthrough of 0.86; the benchmark at $\beta_{\text{exiter}} = 1$ returns the survivor estimate. Taking $\beta_{\text{exiter}} = 0.5$ as a neutral midpoint, the aggregate is 0.93; at three-quarters, 0.97. Under the conservative screen the same calibrations give roughly 0.95 at the floor and 0.975 at the midpoint, tighter but in the same direction.

Table 2: Implied Cohort Passthrough Under Calibrated β_{exiter}

	β_{exiter}					
	0.00	0.25	0.50	0.75	0.90	1.00
Aggregate	0.86	0.90	0.93	0.97	0.99	1.00
Capital	0.82	0.87	0.91	0.96	0.98	1.00
Intermediate	0.79	0.84	0.90	0.95	0.98	1.00
Consumption	0.96	0.97	0.98	0.99	1.00	1.00

Notes: Implied cohort passthrough $\beta_{\text{cohort}} = (1-\theta)\beta_{\text{survivor}} + \theta\beta_{\text{exiter}}$, with $\beta_{\text{survivor}} = 1$ and θ the 2023 value-weighted θ_{high} by end-use. The $\beta_{\text{exiter}} = 0$ column is the zero-exiter floor; $\beta_{\text{exiter}} = 1$ recovers the survivor benchmark. The conservative screen yields tighter bounds in the same direction.

The correction is larger where exit concentrates. Consumption goods barely reroute and stay within a few points of one at every calibration, in line with the near-complete passthrough Flaaen, Hortaçsu, and Tintelnot (2020) and Cavallo et al. (2021) find for consumer goods. Capital and intermediate goods fall to 0.91 and 0.90 at the midpoint, and to 0.82 and 0.79 at the floor. Intermediates and capital reroute most, with exit shares near 0.21 and 0.18 against 0.04 for consumption, so the correction is largest where exit is largest, though it runs ten to twenty points and never overturns the finding that passthrough was high. The same exit that corrects incidence enters welfare twice, through this numerator and through the denominator of the marginal value of public funds. I take the denominator next.

6.2 The Exit-Based MVPF

With β_{cohort} bounded and the inputs to η assembled, I evaluate the MVPF of the 2018 tariffs. Under a nationalist welfare specification, which sets welfare weights on Chinese exporters equal to zero

recovered from how survivor passthrough shifts with the exit share, but $\partial\beta_{\text{survivor}}/\partial\theta$ is estimated too imprecisely to pin β_{marginal}.

and applies a uniform weight on U.S. imports, equation (8) specializes to

$$\mathrm{MVPF}_{\mathrm{exit}} = \frac{\beta_{\mathrm{cohort}}}{(1-\theta)(1+\eta)}. \tag{13}$$

The exit elasticity ψ entering η is recovered from the Imbens–Manski bounds on $\partial\theta/\partial\Delta$ of Section 5.1.2. I evaluate τ and θ at their 2023 values for each end-use category, weighted by 2017 import shares.

Table 3 reports the MVPF by end-use at 2023 exit shares. The low and high columns bracket each estimate, pairing the conservative screen with the full incidence correction ($\beta_{\mathrm{exiter}} = 0$) at one end and the liberal screen with no correction ($\beta_{\mathrm{exiter}} = 1$) at the other. Because each end moves both inputs at once, the two are far apart, so I read the result off the midpoint at $\beta_{\mathrm{exiter}} = 0.5$. Consumption goods barely reroute and their MVPF is 1.07, essentially at the lump-sum benchmark. Capital and intermediate goods, where rerouting concentrates, are higher: 1.17 for capital and 1.39 for intermediates. The aggregate is 1.18, with low–high bounds of 1.03–1.52. No cell crosses into the Laffer region.

Table 3: Marginal Value of Public Funds by End-Use Category (2023)

	Exit share		Fiscal externality		MVPF		
End-use	$\bar{\theta}_{\mathrm{low}}$	$\bar{\theta}_{\mathrm{high}}$	η_{exit} (low)	η_{exit} (high)	MVPF (low)	MVPF (mid)	MVPF (high)
Capital	0.059	0.179	-0.01	-0.25	1.01	1.17	1.62
Consumption	0.014	0.040	-0.01	-0.11	1.01	1.07	1.17
Intermediate	0.068	0.208	-0.06	-0.47	1.06	1.39	2.36
Aggregate	0.045	0.137	-0.03	-0.24	1.03	1.18	1.52

Notes: $\bar{\theta}$ is the 2017-share-weighted rerouted share of baseline trade, and $\eta_{\mathrm{exit}} = -(\bar{\tau}/(1+\bar{\tau}))(\partial\theta/\partial\Delta)/(1-\bar{\theta})$ is the fiscal externality from exit, with $\bar{\tau}$ the 2023 direct tariff weighted by 2017 import shares and $\partial\theta/\partial\Delta$ the Imbens–Manski slopes from Figure 3. The low and high columns pair the conservative screen with full incidence correction ($\beta_{\mathrm{exiter}} = 0$) and the liberal screen with none ($\beta_{\mathrm{exiter}} = 1$); the midpoint takes $\beta_{\mathrm{exiter}} = 0.5$, θ at the screen mean, and the OLS midpoint slope.

Other estimated MVPFs are in a similar range. For example, Finkelstein and Hendren (2020) estimate an MVPF of 2.05, but note that because the change was not marginal, that estimate is too large. Their adjusted MVPF, together with Jaccard (2021), places the range of MVPFs for the trade war between 1.2 and 1.6, not far from an income tax. Applying my incidence correction reduces passthrough below the benchmark of one, so the floor of that range falls toward one while the upper end (where passthrough is already at the benchmark and no correction applies) stays near 1.6. My own exit-based estimate sits in the same territory: a corrected aggregate of 1.18, within a conservative–liberal envelope of 1.03–1.52, despite the fact that mine only accounts for exit.[13]

7 Conclusion

This paper generalizes the canonical textbook approach to tax incidence to environments in which incidence is heterogeneous and agents have the option to exit. In such cases, measured tax incidence may be either biased upward or downward depending on the covariance between exit and incidence. Either feature alone leaves the standard estimator unbiased: heterogeneity without exit recovers the average passthrough across all sellers, and exit under homogeneous passthrough leaves no gap between those who remain and those who leave. The bias arises from the interaction. When agents with systematically different passthrough from the survivors exit the base, the average passthrough among survivors—the object identified in the data—differs from the average across all agents originally affected by the tax. The size of the gap is governed by the share that exits, in the spirit of Chetty (2009b): knowing the exit share and a sign restriction on who exits is enough to bound the correction, without identifying which agents avoided or estimating the structural parameters that drove them. The direction of the bias depends on which agents leave. Section 3 illustrates the flexibility in both directions in the context of an import tariff: when avoidance dominates, the firms that leave are low-cost, low-passthrough sellers, and the standard estimate overstates the burden on domestic buyers; when participation exit dominates, the firms that leave are high-cost, high-passthrough sellers, and the bias flips. The correction is the same in both cases.

While I apply the framework to the 2018 trade war and find a modest correction to the bounds on prevailing estimates on both domestic tariff passthrough and welfare, the range of applications spans public finance. The clearest analog is the personal income tax, where top earners respond to high marginal rates by migrating to lower-tax jurisdictions (Kleven et al. 2020; Rauh and Shyu 2024). The same instrument operates differently in lower-income settings, where small firms and low-skill workers move into the informal sector to escape the tax (Ulyssea 2020). At the same time, corporate income taxes can lead to substantial exit either via shifting paper profits abroad (Tørsløv, Wier, and Zucman 2023) or by literally moving firms (Giroud and Rauh 2019). The exit channel varies, but the bias structure does not. In each case, the gap between the observed and cohort response parameter is governed by the same identity in Proposition 1.

13. Because the tariffs were large and discrete, two factors adjust the welfare cost relative to the marginal MVPF. Adjusting for the demand response over the size of the change lowers it, as in Finkelstein and Hendren (2020). Accounting for the resource costs of avoidance raises it to the extent those costs are real rather than transfers.

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A Theoretical Appendix

The theoretical appendix is organized in three parts. The first proves the framework's results from Section 2, which apply across microfoundations. The second develops the structural model from Section 3, including assumptions, comparative statics, and robustness extensions. The third derives the MVPF formula from Section 6.2.

A.1 Appendix to Section 2

A.1.1 Proof of Proposition 1

Proof. By the definitions of the cohort and survivor estimands,

$$\beta_{\text{survivor}} - \beta_{\text{cohort}} = \frac{\mathbb{E}_\omega[\beta_i\, \sigma_i]}{\mathbb{E}_\omega[\sigma_i]} - \mathbb{E}_\omega[\beta_i].$$

Using the covariance identity $\text{Cov}_\omega(\beta_i, \sigma_i) = \mathbb{E}_\omega[\beta_i\, \sigma_i] - \mathbb{E}_\omega[\beta_i]\mathbb{E}_\omega[\sigma_i]$,

$$\beta_{\text{survivor}} - \beta_{\text{cohort}} = \frac{\text{Cov}_\omega(\beta_i, \sigma_i)}{\mathbb{E}_\omega[\sigma_i]}. \qquad \square$$

A.1.2 Proof of Proposition 2

Proof. Define $\bar{\beta}_{\text{exiter}} \equiv \mathbb{E}_\omega[\beta_i \mid \sigma_i = 0]$. By the law of total expectation,

$$\beta_{\text{cohort}} = (1-\theta)\, \beta_{\text{survivor}} + \theta\, \bar{\beta}_{\text{exiter}}, \tag{A.1}$$

where $\theta = 1 - \mathbb{E}_\omega[\sigma_i]$.

Case (i): $\text{Cov}_\omega(\beta_i, \sigma_i) \geq 0$**.** By the wedge identity, $\beta_{\text{survivor}} \geq \beta_{\text{cohort}}$. Substituting (A.1) gives $\beta_{\text{survivor}} \geq (1-\theta)\beta_{\text{survivor}} + \theta\bar{\beta}_{\text{exiter}}$, which simplifies to $\bar{\beta}_{\text{exiter}} \leq \beta_{\text{survivor}}$. The assumption $\beta_i \geq \beta_{\min}$ implies $\bar{\beta}_{\text{exiter}} \geq \beta_{\min}$. Substituting these inequalities into (A.1),

$$(1-\theta)\, \beta_{\text{survivor}} + \theta\, \beta_{\min} \leq \beta_{\text{cohort}} \leq \beta_{\text{survivor}}.$$

Case (ii): $\text{Cov}_\omega(\beta_i, \sigma_i) \leq 0$**.** The wedge identity gives $\beta_{\text{survivor}} \leq \beta_{\text{cohort}}$, so by the same argument $\bar{\beta}_{\text{exiter}} \geq \beta_{\text{survivor}}$. The assumption $\beta_i \leq \beta_{\max}$ implies $\bar{\beta}_{\text{exiter}} \leq \beta_{\max}$. Substituting,

$$\beta_{\text{survivor}} \leq \beta_{\text{cohort}} \leq (1-\theta)\, \beta_{\text{survivor}} + \theta\, \beta_{\max}. \qquad \square$$

A.1.3 Disaggregated Bound Under Margin-Specific Selection

When the global covariance $\mathrm{Cov}_\omega(\beta_i, \sigma_i)$ has ambiguous sign—for instance, when offsetting exit margins push the wedge in opposite directions and approximately cancel—Proposition 2 delivers no informative bound. The decomposition following Proposition 1 permits a margin-specific bound when the sign of $\beta_{\text{survivor}} - \bar{\beta}_k$ is known for each k.

Partition the exit margins into $K^+ \equiv \{k : \bar{\beta}_k < \beta_{\text{survivor}}\}$ and $K^- \equiv \{k : \bar{\beta}_k > \beta_{\text{survivor}}\}$, with shares $\theta^+ \equiv \sum_{k \in K^+} \theta_k$ and $\theta^- \equiv \sum_{k \in K^-} \theta_k$.

Corollary 3 (Disaggregated bound)**.** *Suppose Assumption 1(i) holds, exit decomposes into K mutually exclusive margins as in equation (6), and the sign of $\beta_{survivor} - \bar{\beta}_k$ is known for each k. Then*

$$\beta_{cohort} \in \left[(1-\theta^+)\,\beta_{survivor} + \theta^+\,\beta_{\min},\ (1-\theta^-)\,\beta_{survivor} + \theta^-\,\beta_{\max}\right].$$

Proof. From the cohort decomposition $\beta_{\text{cohort}} = (1-\theta)\beta_{\text{survivor}} + \sum_k \theta_k \bar{\beta}_k$, the constraints $\bar{\beta}_k \in [\beta_{\min}, \beta_{\text{survivor}}]$ for $k \in K^+$ and $\bar{\beta}_k \in [\beta_{\text{survivor}}, \beta_{\max}]$ for $k \in K^-$ deliver the stated interval at the extreme values. The lower endpoint sets $\bar{\beta}_k = \beta_{\min}$ on K^+ and $\bar{\beta}_k = \beta_{\text{survivor}}$ on K^-, giving $(1-\theta)\beta_{\text{survivor}} + \theta^+\beta_{\min} + \theta^-\beta_{\text{survivor}} = (1-\theta^+)\beta_{\text{survivor}} + \theta^+\beta_{\min}$. The upper endpoint is symmetric. □

The interval nests the cases of Proposition 2: when $K^- = \emptyset$, $\theta^+ = \theta$ and the lower endpoint reduces to $(1-\theta)\beta_{\text{survivor}} + \theta\beta_{\min}$ as in case (i); when $K^+ = \emptyset$, the upper endpoint reduces to case (ii). When both sets are nonempty, the interval contains β_{survivor} in the interior, reflecting the wider identified set under offsetting margins. The result tightens the worst-case Manski bound uniformly: replacing θ with the smaller θ^+ on the lower endpoint and with the smaller θ^- on the upper endpoint shrinks the interval on both sides whenever both margin classes are nonempty. The informational requirement is stronger than Proposition 2: rather than a single sign on the global covariance, the analyst must know each margin's selection direction, typically delivered by the same theoretical structure that signs the global covariance margin-by-margin, as in Corollary 2.

A.1.4 Proof of Corollary 1

Proof. Let ξ_i have continuous density $f > 0$ on its support and CDF F. Without loss of generality take $\beta(\cdot)$ monotone increasing in ξ (the decreasing case is symmetric).

Case (i): low types exit. Under threshold selection with low types exiting, $\sigma_i = \mathbf{1}\{\xi_i \geq \xi^*(\theta)\}$ where the cutoff satisfies $\theta = F(\xi^*(\theta))$. Hence $d\xi^*/d\theta = 1/f(\xi^*(\theta))$.

Define

$$G(\theta) \equiv (1-\theta)\,\beta_{\text{survivor}}(\theta) = \int_{\xi^*(\theta)}^{\infty} \beta(\xi)\,f(\xi)\,d\xi.$$

Differentiating the integral form,

$$\frac{dG}{d\theta} = -\beta(\xi^*(\theta))\, f(\xi^*(\theta))\, \frac{d\xi^*}{d\theta} = -\beta(\xi^*(\theta)).$$

Differentiating the product form,

$$\frac{dG}{d\theta} = -\beta_{\text{survivor}}(\theta) + (1-\theta)\, \frac{\partial \beta_{\text{survivor}}}{\partial \theta}.$$

Equating the two expressions,

$$\beta(\xi^*(\theta)) = \beta_{\text{survivor}}(\theta) - (1-\theta)\, \frac{\partial \beta_{\text{survivor}}}{\partial \theta} \equiv \beta_{\text{marginal}}(\theta).$$

Survivors satisfy $\xi_i \geq \xi^*(\theta)$. Monotonicity of β gives $\beta(\xi_i) \geq \beta(\xi^*(\theta))$ for all survivors, so

$$\beta_{\text{survivor}}(\theta) = \mathbb{E}[\beta(\xi) \mid \xi \geq \xi^*(\theta)] \geq \beta(\xi^*(\theta)) = \beta_{\text{marginal}}(\theta),$$

with strict inequality whenever β is non-constant on the survivor set.

Exiters satisfy $\xi_i < \xi^*(\theta)$, so monotonicity gives $\beta(\xi_i) \leq \beta_{\text{marginal}}(\theta)$. Combined with $\beta_i \geq \beta_{\min}$,

$$\bar{\beta}_{\text{exiter}} \equiv \mathbb{E}_\omega[\beta_i \mid \sigma_i = 0] \in [\beta_{\min}, \beta_{\text{marginal}}(\theta)].$$

Substituting into the law of total expectation $\beta_{\text{cohort}} = (1-\theta)\, \beta_{\text{survivor}} + \theta\, \bar{\beta}_{\text{exiter}}$ yields

$$\beta_{\text{cohort}} \in \big[(1-\theta)\, \beta_{\text{survivor}} + \theta\, \beta_{\min},\ (1-\theta)\, \beta_{\text{survivor}} + \theta\, \beta_{\text{marginal}}\big].$$

This interval is contained in the Proposition 2 case (i) interval $[(1-\theta)\, \beta_{\text{survivor}} + \theta\, \beta_{\min}, \beta_{\text{survivor}}]$, with strict containment whenever $\beta_{\text{marginal}} < \beta_{\text{survivor}}$.

Case (ii): high types exit. Under threshold selection with high types exiting, $\sigma_i = \mathbf{1}\{\xi_i \leq \bar{\xi}(\theta)\}$ where $\theta = 1 - F(\bar{\xi}(\theta))$. Hence $d\bar{\xi}/d\theta = -1/f(\bar{\xi}(\theta))$.

Define $G(\theta) = (1-\theta)\, \beta_{\text{survivor}}(\theta) = \int_{-\infty}^{\bar{\xi}(\theta)} \beta(\xi)\, f(\xi)\, d\xi$. Differentiating two ways,

$$\frac{dG}{d\theta} = \beta(\bar{\xi}(\theta))\, f(\bar{\xi}(\theta))\, \frac{d\bar{\xi}}{d\theta} = -\beta(\bar{\xi}(\theta)),$$

$$\frac{dG}{d\theta} = -\beta_{\text{survivor}}(\theta) + (1-\theta)\, \frac{\partial \beta_{\text{survivor}}}{\partial \theta}.$$

Equating gives the same formula,

$$\beta(\bar{\xi}(\theta)) = \beta_{\text{survivor}}(\theta) - (1-\theta)\, \frac{\partial \beta_{\text{survivor}}}{\partial \theta} = \beta_{\text{marginal}}(\theta).$$

Survivors satisfy $\xi_i \leq \bar{\xi}(\theta)$, so monotonicity gives $\beta_{\text{survivor}} \leq \beta_{\text{marginal}}$, with strict inequality under non-constant β on the survivor set.

Exiters satisfy $\xi_i > \bar{\xi}(\theta)$, so $\beta(\xi_i) \geq \beta_{\text{marginal}}$. Combined with $\beta_i \leq \beta_{\max}$,

$$\bar{\beta}_{\text{exiter}} \in [\beta_{\text{marginal}}(\theta),\ \beta_{\max}].$$

Substituting into the law of total expectation,

$$\beta_{\text{cohort}} \in \left[(1-\theta)\,\beta_{\text{survivor}} + \theta\,\beta_{\text{marginal}},\ (1-\theta)\,\beta_{\text{survivor}} + \theta\,\beta_{\max}\right],$$

which is contained in the Proposition 2 case (ii) interval $[\beta_{\text{survivor}},\ (1-\theta)\,\beta_{\text{survivor}} + \theta\,\beta_{\max}]$, with strict containment whenever $\beta_{\text{marginal}} > \beta_{\text{survivor}}$. □

A.2 Appendix to Section 3

A.2.1 Assumptions about Demand

I impose several regularity conditions on demand:

Assumption 3 (Demand)**.** *The demand for each* variety i *depends on its own delivered price* $\tilde{p}_i$ *and on a price aggregator* $\mathbf{P}$ *summarizing rivals. Fixing* $\mathbf{P}$ *as parametric at the variety level, the primitives satisfy:*

D1. Downward sloping own demand and outward shifts in rivals: $\dfrac{\partial D_i}{\partial \tilde{p}_i} < 0$ *and* $\dfrac{\partial D_i}{\partial \mathbf{P}} > 0$.

D2. Smoothness: $D_i(\cdot \mid \mathbf{P}) \in C^2$ *in own price.*

D3. Elastic demand: $\varepsilon_i(\tilde{p}_i \mid \mathbf{P}) \equiv -\dfrac{\tilde{p}_i}{D_i}\dfrac{\partial D_i}{\partial \tilde{p}_i} > 1$.

D4. Profit concavity / local stability: *For either channel* $c \in \{d, a\}$ *with delivered price* $\tilde{p}_i^c$*,*

$$\varepsilon(\tilde{p}_i^c \mid \mathbf{P}) \ - \ 1 \ - \ \kappa(\tilde{p}_i^c \mid \mathbf{P}) \ > \ 0.$$

D5. Kimball curvature: $\kappa_i(\tilde{p}_i \mid \mathbf{P}) \equiv -\dfrac{\partial \ln \varepsilon_i(\tilde{p}_i \mid \mathbf{P})}{\partial \ln \tilde{p}_i} \leq 0$ *(equals* 0 *under CES). We also impose two extra conditions:*

(a) $\dfrac{\partial \kappa(\tilde{p}_i \mid \mathbf{P})}{\partial \ln \tilde{p}_i} \leq 0.$

(b) $\kappa'(\tilde{p}) \ \equiv \partial\kappa(\tilde{p})/\partial \ln \tilde{p} \geq \ -\frac{\varepsilon(\tilde{p})}{\varepsilon(\tilde{p})-1}\,\kappa(\tilde{p})^2.$

Conditions **D1**–**D4** are standard and ensure well-behaved monopolistic competition with elastic demand and unique profit-maximizing prices. Condition **D5** introduces Kimball-type demand, where $\kappa \leq 0$ means the demand elasticity ε is weakly increasing in price, and the technical sub-conditions **D5(a)–(b)** ensure that the resulting passthrough is monotone in marginal cost. The empirical literature consistently finds heterogeneous markups across firms (Gopinath and Itskhoki 2010; Amiti, Itskhoki, and Konings 2019; De Loecker, Eeckhout, and Unger 2020; Edmond, Midrigan, and Xu 2023), of which Kimball demand is a tractable microfoundation.

A.2.2 Proof of Lerner Condition

Proof. Interior and FOC. By **D2**–**D4**, $\pi_i(\cdot)$ is strictly concave in p_i. The unique maximizer is interior and characterized by the FOC. Let the delivered price be

$$\tilde{p}_i \equiv \tilde{p}_i(p_i, \tau^r) = \begin{cases} (1+\tau^a)p_i & \text{(rerouting channel)} \\ (1+\tau^d)\,p_i & \text{(direct channel).} \end{cases}$$

Then $\partial \tilde{p}_i/\partial p_i \in \{1+\tau^a,\ 1+\tau^d\}$, and the FOC is

$$0 = \frac{\partial \pi_i}{\partial p_i} = D_i(\tilde{p}_i) + (p_i - m_i)\, D_i'(\tilde{p}_i)\,\frac{\partial \tilde{p}_i}{\partial p_i}.$$

With $\varepsilon_i(\tilde{p}_i) \equiv -\tilde{p}_i D_i'(\tilde{p}_i)/D_i(\tilde{p}_i)$,

$$\frac{p_i^* - m_i}{p_i^*} = \frac{\tilde{p}_i}{(\partial \tilde{p}_i/\partial p_i)\, p_i^*} \cdot \frac{-D_i(\tilde{p}_i)}{\tilde{p}_i D_i'(\tilde{p}_i)} = \frac{1}{\varepsilon_i(\tilde{p}_i)}.$$

This is the Lerner condition.

Monotonicity in m_i. Define

$$g(p, m; \tau^r) \equiv D(\tilde{p}) + (p - m)\, D'(\tilde{p})\,\frac{\partial \tilde{p}}{\partial p}, \qquad \tilde{p} = \tilde{p}(p, \tau^r).$$

At the optimum, $g(p^*, m; \tau^r) = 0$. Partials:

$$\frac{\partial g}{\partial m} = -\,D'(\tilde{p})\,\frac{\partial \tilde{p}}{\partial p} > 0 \quad (\mathbf{D1}), \qquad \frac{\partial g}{\partial p} = 2\,D'(\tilde{p})\,\frac{\partial \tilde{p}}{\partial p} + (p-m)\,D''(\tilde{p})\left(\frac{\partial \tilde{p}}{\partial p}\right)^2.$$

Strict concavity of profits (**D4**) implies $\partial g/\partial p < 0$ at p^*. By the implicit function theorem, $\partial p^*/\partial m = -(\partial g/\partial m)/(\partial g/\partial p) > 0$.

Monotonicity in τ^r. For the direct channel, $\tilde{p} = (1+\tau^r)p$. Write

$$G(p, \tau^r) \;\equiv\; \frac{p-m}{p} - \frac{1}{\varepsilon((1+\tau^r)p)} \;=\; 0.$$

Then

$$\frac{\partial G}{\partial \tau^r} = \frac{\varepsilon'((1+\tau^r)p)}{\varepsilon((1+\tau^r)p)^2}\, p \;\geq 0, \qquad \frac{\partial G}{\partial p} = \frac{m}{p^2} + \frac{\varepsilon'((1+\tau^r)p)}{\varepsilon((1+\tau^r)p)^2}\,(1+\tau^r).$$

Under Kimball demand, $\varepsilon'(\cdot) \geq 0$, so $\partial G/\partial \tau^r \geq 0$. Profit concavity (D4, equivalently $\varepsilon - 1 - \kappa > 0$ at the optimum) implies $\partial G/\partial p > 0$. Hence, by the IFT,

$$\frac{\partial p^*}{\partial \tau^r} = -\frac{\partial G/\partial \tau^r}{\partial G/\partial p} \;\leq\; 0,$$

with equality under CES ($\varepsilon' \equiv 0$).

Quantities. In either channel $\tilde{p} = (1+\tau^r)p^*$ and

$$\frac{d\big((1+\tau^r)p^*\big)}{d\tau^r} = p^* + (1+\tau^r)\frac{dp^*}{d\tau^r} = p^*\left(1 + \frac{\kappa(\tilde{p})}{\varepsilon(\tilde{p}) - 1 - \kappa(\tilde{p})}\right) = p^*\frac{\varepsilon(\tilde{p}) - 1}{\varepsilon(\tilde{p}) - 1 - \kappa(\tilde{p})} \;>\; 0,$$

since $\varepsilon(\tilde{p}) > 1$ and the denominator is positive by D4. With $D'(\cdot) < 0$, $q_i^* = D(\tilde{p})$ strictly falls in τ^r. Similarly, q_i^* strictly falls in m because p^* rises in m and demand slopes down. This proves (i)–(iii). □

Lemma 1 establishes two key results. First, it confirms the standard Lerner condition. Firms set their markup inversely to the elasticity of demand they face. This makes all comparative statics run through how a given wedge, τ^r, shifts the delivered price and how elasticity varies with that price. Second, the lemma establishes the monotone comparative statics that are essential for the paper's sorting mechanism.

Part (i) shows that a higher marginal cost pushes a firm's optimal pre-wedge price upward, providing the clean ordering in m necessary to derive a unique sorting cutoff. Part (ii) highlights that any wedge acts as a demand shifter. Part (iii) records the quantity implications. A higher marginal cost or a higher wedge τ^r reduces the quantity sold. This decline in profitability within a channel is what ultimately drives selection when wedges differ, a mechanism formalized next.

A.2.3 Proof of Proposition 3

Proof. Write the first-order condition for a firm in the direct channel as $G(p, \tau^d) \equiv (p-m)/p - 1/\varepsilon((1+\tau^d)p) = 0$. Totally differentiate and use

$$\frac{\partial G}{\partial \tau^d} = \frac{\varepsilon'((1+\tau^d)p)}{\varepsilon((1+\tau^d)p)^2}\, p, \qquad \frac{\partial G}{\partial p} = \frac{m}{p^2} + \frac{\varepsilon'((1+\tau^d)p)}{\varepsilon((1+\tau^d)p)^2}\,(1+\tau^d).$$

Using $\kappa(u) = -u\,\varepsilon'(u)/\varepsilon(u)$ and the Lerner condition $m = p^*(\varepsilon-1)/\varepsilon$,

$$\frac{dp^*}{d\tau^d} = -\frac{\partial G/\partial \tau^d}{\partial G/\partial p} = \frac{\kappa(\tilde{p}^d)}{1+\tau^d} \cdot \frac{p^*}{\varepsilon(\tilde{p}^d) - 1 - \kappa(\tilde{p}^d)}.$$

Finally, the passthrough elasticity is

$$\beta_i = \frac{d\ln\big((1+\tau^d)p^*\big)}{d\tau^d} = \frac{1}{1+\tau^d} + \frac{1}{p^*}\frac{dp^*}{d\tau^d} = \frac{1}{1+\tau^d} \cdot \frac{\varepsilon(\tilde{p}^d) - 1}{\varepsilon(\tilde{p}^d) - 1 - \kappa(\tilde{p}^d)}.$$

Under Assumption **D4**, strict profit concavity at the optimum is equivalent to $\varepsilon - 1 - \kappa > 0$, which ensures the denominator is positive. Differentiating $\beta = (1+\tau^d)^{-1}(\varepsilon-1)/(\varepsilon-1-\kappa)$ in $\ln\tilde{p}$ and using $\partial\varepsilon/\partial\ln\tilde{p} = -\kappa\varepsilon$,

$$\frac{\partial \beta}{\partial \ln \tilde{p}} = \frac{1}{1+\tau^d} \cdot \frac{\kappa^2\,\varepsilon \;+\; (\varepsilon-1)\,\kappa'}{(\varepsilon-1-\kappa)^2}.$$

Under **D5(b)**, $(\varepsilon-1)\kappa' \geq -\varepsilon\kappa^2$, so the numerator is ≥ 0, with strict inequality whenever $\kappa < 0$. Combined with Lemma 1, which gives $\partial\tilde{p}/\partial m > 0$, this yields $\partial\beta/\partial m \geq 0$, strict under Kimball ($\kappa < 0$) and $= 0$ under CES ($\kappa \equiv 0$). □

A.2.4 Proof of Proposition 4

Proof. By **D2**–**D4**, the maximizer in $V(m, 1+\tau^r)$ is interior and unique for each $r \in \{d, a\}$; envelope arguments apply.

Monotonicity of ΔV. The primitive $(p-m)D((1+\tau^r)p)$ has cross-partial in $(m, 1+\tau^r)$ equal to $-p\,D'((1+\tau^r)p) > 0$ by **D1**, so it has strictly increasing differences in $(m, 1+\tau^r)$. By Topkis, V inherits strictly increasing differences, so $V_m(m, 1+\tau^r)$ is strictly increasing in $1+\tau^r$. Since $\tau^a < \tau^d$,

$$\Delta V_m(m) \equiv V_m(m, 1+\tau^a) - V_m(m, 1+\tau^d) < 0,$$

so ΔV is strictly decreasing in m.

Existence and uniqueness of cutoffs. The avoidance cutoff solves $\Delta V(\widehat{m}) = F_a$. Since ΔV is continuous and strictly decreasing, a unique $\widehat{m}$ satisfies the indifference condition whenever $F_a \in [\Delta V(\bar{m}), \Delta V(\underline{m})]$ (the interior case). The participation cutoff solves $V(\bar{m}, 1+\tau^d) = F_d$.

By envelope, $V_m(m, 1+\tau^d) = -q^d(m) < 0$, so $V(\cdot, 1+\tau^d)$ is continuous and strictly decreasing, giving a unique $\bar{m}$ for any $F_d \in (0, V(\underline{m}, 1+\tau^d))$.

Ordering. By Assumption 2, $F_a \geq \Delta V(\bar{m})$. Combined with $\Delta V(\widehat{m}) = F_a$ and strict monotonicity of ΔV, $\widehat{m} \leq \bar{m}$.

Partition. Let $\pi_d(m) = V(m, 1+\tau^d) - F_d$, $\pi_a(m) = V(m, 1+\tau^a) - F_a - F_d$, and $\pi_e = 0$.

For $m < \widehat{m}$: from $\widehat{m} \leq \bar{m}$ and strict monotonicity of $V(\cdot, 1+\tau^d)$, $V(m, 1+\tau^d) > F_d$, so $\pi_d(m) > 0$; and $\Delta V(m) > \Delta V(\widehat{m}) = F_a$ gives $\pi_a(m) - \pi_d(m) = \Delta V(m) - F_a > 0$. Hence $\pi_a > \pi_d > 0 = \pi_e$, and reroute is optimal.

For $\widehat{m} \leq m \leq \bar{m}$: $\Delta V(m) \leq F_a$ gives $\pi_d \geq \pi_a$, and $V(m, 1+\tau^d) \geq F_d$ gives $\pi_d \geq 0 = \pi_e$. Direct is optimal.

For $m > \bar{m}$: $V(m, 1+\tau^d) < F_d$ gives $\pi_d < 0$. By Assumption 2 and strict monotonicity of ΔV, $\Delta V(m) < \Delta V(\bar{m}) \leq F_a$, so $V(m, 1+\tau^a) < V(m, 1+\tau^d) + F_a < F_d + F_a$, hence $\pi_a(m) < 0$. Exit is optimal.

Boundary firms are indifferent on a measure-zero set under continuous G, so the partition holds almost surely.

Comparative statics. By the implicit function theorem applied to $\Delta V(\widehat{m}) = F_a$ and $V(\bar{m}, 1+\tau^d) = F_d$:

$$\frac{\partial \widehat{m}}{\partial \tau^d} = -\frac{\partial \Delta V/\partial \tau^d}{\Delta V_m}, \quad \frac{\partial \widehat{m}}{\partial \tau^a} = -\frac{\partial \Delta V/\partial \tau^a}{\Delta V_m}, \quad \frac{\partial \bar{m}}{\partial \tau^d} = -\frac{\partial V/\partial \tau^d}{V_m}, \quad \frac{\partial \bar{m}}{\partial F_d} = \frac{1}{V_m}.$$

By envelope, $\partial V(m, 1+\tau^r)/\partial \tau^r = (p^* - m)\, p^*\, D'((1+\tau^r)p^*) < 0$, since $p^* > m$ (Lemma 1) and $D' < 0$ (**D1**). Hence $\partial \Delta V/\partial \tau^d = -\partial V(m, 1+\tau^d)/\partial \tau^d > 0$ and $\partial \Delta V/\partial \tau^a = \partial V(m, 1+\tau^a)/\partial \tau^a < 0$. Combined with $\Delta V_m < 0$ and $V_m < 0$, the four signs in the proposition follow. □

A.2.5 Proof of Corollary 2

Proof. Specialize the decomposition (6) of Section 2 to $K = 2$ exit margins, $k \in \{a, p\}$, with $\theta_a = \Pr(m < \widehat{m})$, $\theta_p = \Pr(m > \bar{m})$, $\bar{\beta}_a \equiv \mathbb{E}[\beta(m) \mid m < \widehat{m}]$, $\bar{\beta}_p \equiv \mathbb{E}[\beta(m) \mid m > \bar{m}]$, and $\beta_{\text{survivor}} = \mathbb{E}[\beta(m) \mid \widehat{m} \leq m \leq \bar{m}]$ from Proposition 4. Then

$$\beta_{\text{survivor}} - \beta_{\text{cohort}} = \theta_a\,(\beta_{\text{survivor}} - \bar{\beta}_a) + \theta_p\,(\beta_{\text{survivor}} - \bar{\beta}_p). \tag{A.2}$$

Sign of each term. By Proposition 3, low-cost firms have lower passthrough under Kimball, so $\bar{\beta}_a < \beta_{\text{survivor}}$ and the avoidance contribution $\theta_a(\beta_{\text{survivor}} - \bar{\beta}_a)$ is positive. High-cost firms have higher passthrough, so $\bar{\beta}_p > \beta_{\text{survivor}}$ and the participation contribution $\theta_p(\beta_{\text{survivor}} - \bar{\beta}_p)$ is negative.

Dominance. The survivor-cohort gap is positive, $\beta_{\text{survivor}} > \beta_{\text{cohort}}$, if and only if the magnitude of the positive contribution exceeds that of the negative contribution:

$$\theta_a\,(\beta_{\text{survivor}} - \bar{\beta}_a) \;>\; \theta_p\,(\bar{\beta}_p - \beta_{\text{survivor}}),$$

which is equation (11) in the body. Two special cases follow. If $\theta_p = 0$ (no participation exit), the avoidance margin dominates trivially. If $\theta_a = \theta_p$ and the within-region passthrough gaps are equal, the two margins offset exactly and $\beta_{\text{survivor}} = \beta_{\text{cohort}}$. □

A.3 MVPF Derivation

This appendix derives equation (8). The policy variable is the tax rate τ at which revenue is collected on the observed channel.

The planner chooses τ to maximize welfare subject to a revenue constraint:

$$\max_{\tau} W(\tau) \quad \text{subject to} \quad G = R(\tau). \tag{SPP}$$

The MVPF is the shadow value of relaxing the constraint, MVPF = $(-\partial W/\partial\tau)/(\partial R/\partial\tau)$.

By the envelope theorem (extended to the exit margin via revealed preference; see Section 2.3), the first-order welfare loss per unit pre-tax base is $\beta_i\,d\tau$ on buyers and $(1-\beta_i)\,d\tau$ on sellers, with equality for survivors and an upper bound for exiters. Aggregating under uniform within-group planner weights λ_B and λ_S,

$$-\frac{\partial W}{\partial \tau} \;\le\; B^{\text{cohort}}\left[\lambda_B\,\beta_{\text{cohort}} + \lambda_S\,(1-\beta_{\text{cohort}})\right],$$

where $\beta_{\text{cohort}} = (1-\theta)\,\beta_{\text{survivor}} + \theta\,\bar{\beta}_{\text{exiter}}$.

Revenue on the surviving base is $R(\tau) = \tau\,(1-\theta(\tau))\,B^{\text{cohort}}$. Differentiating in τ,

$$\frac{\partial R}{\partial \tau} \;=\; (1-\theta)\,B^{\text{cohort}} \;-\; \tau\,\frac{\partial \theta}{\partial \tau}\,B^{\text{cohort}}.$$

With $\psi \equiv \partial \ln\theta/\partial\ln\tau$ the elasticity of the exit share with respect to the tax rate, $\tau\,\partial\theta/\partial\tau = \theta\,\psi$, so

$$\frac{\partial R}{\partial \tau} \;=\; B^{\text{cohort}}\,(1-\theta)\left[1 - \frac{\theta}{1-\theta}\,\psi\right] \;=\; B^{\text{cohort}}\,(1-\theta)\,(1+\eta),$$

with $\eta = -(\theta/(1-\theta))\,\psi$. Equivalently, $\eta = \partial\ln(1-\theta)/\partial\ln\tau$ is the elasticity of the surviving base with respect to the tax rate; $\eta < 0$ whenever exit rises with the tax, and the denominator vanishes at $\eta = -1$.

Taking the ratio,

$$\text{MVPF} \;=\; \frac{\lambda_B\,\beta_{\text{cohort}} + \lambda_S\,(1-\beta_{\text{cohort}})}{(1-\theta)\,(1+\eta)},$$

which is equation (8); B^{cohort} cancels.

The regression of Section 5.1.2 identifies the wedge-derivative $\partial\theta/\partial\Delta$, with $\Delta = \log[(1+\tau)/(1+\tau^a)]$. Holding τ^a fixed, $\partial\Delta/\partial\tau = 1/(1+\tau)$, so the chain rule gives

$$\psi \;=\; \frac{\partial \ln\theta}{\partial \ln\tau} \;=\; \frac{\tau}{(1+\tau)\,\theta}\,\frac{\partial\theta}{\partial\Delta},$$

and equivalently $\eta = -(\tau/((1+\tau)(1-\theta)))\,\partial\theta/\partial\Delta$. Section 6.2 evaluates this expression at the 2017-share-weighted direct tariff and exit share for each end-use category.

When exit decomposes into K mutually exclusive margins as in equation (6),

$$\psi \;=\; \sum_{k=1}^{K} \frac{\theta_k}{\theta}\,\psi_k, \qquad \psi_k \;\equiv\; \frac{\partial \ln\theta_k}{\partial \ln\tau},$$

and the fiscal externality decomposes as

$$\eta \;=\; -\sum_{k=1}^{K} \frac{\theta_k}{1-\theta}\,\psi_k.$$

When one margin dominates the aggregate is approximated by its contribution alone; counterfactual evaluation under composition-shifting policy requires the ψ_k separately.

B Screening Implementation Details

This appendix provides complete documentation of the methodology used to construct the exit share θ. The goal is to identify tariff-induced rerouting of Chinese goods through third countries to the United States.

B.1 Data and Notation

I use annual bilateral trade flows from CEPII's BACI HS-12 vintage (2012–2023) at the HS6 product level. For each year t, country pair (i, j), and HS6 code k, I observe the customs value of shipments $v_{i\to j,k,t}$.

For each HS6 product k and hub i in year t, define the two potential legs of rerouting:

- First leg (China $\longrightarrow$ hub): $\ell_{k,i,t} \equiv v_{\text{CHN}\to i,k,t}$

- Second leg (hub $\longrightarrow$ United States): $x_{k,i,t} \equiv v_{i\to\text{USA},k,t}$

Define market shares as follows. Let $s^{\text{US}}_{k,i,t}$ denote country i's share of U.S. imports in product k:

$$s^{\text{US}}_{k,i,t} = \frac{v_{i\to\text{USA},k,t}}{\sum_j v_{j\to\text{USA},k,t}}.$$

Let $s^{\text{ROW}}_{k,i,t}$ denote country i's share of exports to the rest of the world (excluding the U.S.):

$$s^{\text{ROW}}_{k,i,t} = \frac{\sum_{j\neq\text{USA}} v_{i\to j,k,t}}{\sum_{j\neq\text{USA}}\sum_m v_{m\to j,k,t}}.$$

B.2 Hub Countries

I restrict potential hubs to 36 countries with geographic proximity to China, port infrastructure, and established trade relationships with both China and the U.S. (Table B.1). This excludes implausible routes and reduces false positives from noise in global trade data.

Table B.1: Potential Transshipment Hubs (ISO-3 codes)

ARE	BHR	CAN	CHL	DEU	DOM	EGY	HKG	IDN	IND	ISR	JOR
JPN	KAZ	KHM	LAO	LKA	MAC	MAR	MEX	MYS	NLD	NPL	OMN
PAK	PAN	PHL	POL	RUS	SAU	SGP	THA	TUR	TWN	VNM	ZAF

Note: In BACI, Taiwan is recorded as "Other Asia" with country code S19.

B.3 Trend Estimation

I estimate product- and partner-specific trends in the pre-tariff period (2012–2017) with time centered at 2014.5.

Share trends (logit). For China and hub shares in U.S. imports and in ROW exports:

$$\text{logit}(s_{k,i,t}) = \alpha_{k,i} + \beta_{k,i}(t-2014.5) + \varepsilon_{k,i,t}, \quad \widehat{s}_{k,i,t} = \text{invlogit}(\widehat{\alpha}_{k,i} + \widehat{\beta}_{k,i}(t-2014.5)).$$

If estimation is unreliable (fewer than 3 non-zero observations), I use the pre-period mean $\bar{s}_{k,i,\text{pre}}$.

Level trends (log-linear). For trade levels $v \in \{\ell_{k,i,t}, x_{k,i,t}\}$:

$$\log(\max(v,1)) = a_{k,i} + g_{k,i}(t-2014.5) + \nu_{k,i,t}, \quad \widehat{v}_{k,i,t} = \exp(\widehat{a}_{k,i} + \widehat{g}_{k,i}(t-2014.5)).$$

If estimation is unreliable, I use the pre-period mean $\bar{v}_{k,i,\text{pre}}$.

Surprise growth. Define above-trend growth, floored at zero:

$$\tilde{\ell}_{k,i,t} = \max\{\ell_{k,i,t} - \widehat{\ell}_{k,i,t}, 0\} \quad \text{and} \quad \tilde{x}_{k,i,t} = \max\{x_{k,i,t} - \widehat{x}_{k,i,t}, 0\}.$$

B.4 Screening Criteria

To be flagged as rerouting, a product-hub-year triplet (k, i, t) must simultaneously satisfy five conditions.

Screen 1: Tariff Exposure. Product k must have been exposed to U.S. tariff increases on China:

$$\tau^d_{k,2019} > \tau^d_{k,2017},$$

where $\tau^d_{k,t}$ is the U.S. ad valorem tariff on Chinese product k in year t.

Screen 2: U.S. Reallocation. China's share of U.S. imports must fall below its pre-trend while the hub's share rises above its pre-trend:

$$s^{\text{US}}_{k,\text{CHN},t} < \widehat{s}^{\text{US}}_{k,\text{CHN},t} \quad \text{and} \quad s^{\text{US}}_{k,i,t} > \widehat{s}^{\text{US}}_{k,i,t}.$$

Screen 3: Rest-of-World Specificity. China's ROW export share must remain stable or rise relative to the hub's. Specifically, China's above-trend gain must weakly exceed the hub's above-trend gain:

$$\left(s^{\text{ROW}}_{k,\text{CHN},t} - \widehat{s}^{\text{ROW}}_{k,\text{CHN},t}\right) \geq \gamma \left(s^{\text{ROW}}_{k,i,t} - \widehat{s}^{\text{ROW}}_{k,i,t}\right),$$

where $\gamma = 1$. This rules out cases where China is losing global competitiveness in product k.

Screen 4: Tariff Feasibility. The hub country's tariff on product k must be lower than the U.S. tariff on China:

$$\tau^{\text{hub}}_{k,i,t} < \tau^d_{k,t}.$$

Hub tariffs are MFN rates from Teti (2025). This ensures there is an economic incentive to reroute through hub i.

Screen 5: Quantitative Consistency. The China-to-hub flow must be large enough to plausibly supply the hub-to-U.S. flow. I impose two versions:

(a) **Conservative (growth-based):** Uses above-trend growth on both legs. The condition is:

$$\tilde{\ell}_{k,i,t} \geq \rho^* \max(\tilde{x}_{k,i,t}, x_{\text{hinge},\Delta}).$$

The flagged quantity is the bottleneck: $\min(\tilde{\ell}_{k,i,t}, \tilde{x}_{k,i,t})$.

(b) **Liberal (levels-based):** Uses trade levels directly. The conditions are:

$$\min(\ell_{k,i,t}, x_{k,i,t}) \geq x_{\text{hinge,levels}} \quad \text{and} \quad \ell_{k,i,t} \geq \phi^* x_{k,i,t}.$$

The flagged quantity is the bottleneck: $\min(\ell_{k,i,t}, x_{k,i,t})$.

Parameter Calibration Table B.2 lists fixed parameters and calibrated thresholds.

Table B.2: Screening Parameters

Parameter	Value	Description
γ	1.0	ROW specificity multiplier (Screen 3).
$x_{\text{hinge},\Delta}$	0	Hinge for $\tilde{x}$ in growth rule (thousands of current USD).
$x_{\text{hinge,levels}}$	2000	Minimum size for $\min(\ell, x)$ in levels rule (thousands of current USD).
ρ^*	0	Growth-rule proportionality threshold (Screen 5a).
ϕ^*	0.325	Levels-rule proportionality threshold (Screen 5b).

The calibrated values ρ^* and ϕ^* are selected via placebo discipline. I search over a grid and choose the smallest values such that the implied exit share for never-treated products (those with $\tau^d_{k,2019} = \tau^d_{k,2017}$) does not exceed 1% in any pre-tariff year (2012–2017). This yields $\rho^* = 0$ (any positive above-trend growth on both legs qualifies) and $\phi^* = 0.325$ (the China-to-hub leg must be at least 32.5% of the hub-to-U.S. leg).

Aggregation and Capping For each product k and year t, I aggregate flagged values across all hubs passing Screens 1–5:

$$\theta^{\text{raw}}_{k,t} = \sum_{i \in \mathcal{H}} \mathbf{1}[\text{Screens 1–5 pass for } (k, i, t)] \times q_{k,i,t},$$

where $q_{k,i,t}$ is the flagged quantity from Screen 5 (either $\min(\tilde{\ell}_{k,i,t}, \tilde{x}_{k,i,t})$ for the conservative version or $\min(\ell_{k,i,t}, x_{k,i,t})$ for the liberal version).

I then deflate to 2017 dollars and cap at the 2017 baseline:

$$\theta_{k,t} = \min\left(\frac{\theta^{\text{raw}}_{k,t}/P_t}{v_{\text{CHN}\to\text{USA},k,2017}}, 1\right),$$

where P_t is the BLS end-use import price index for all commodities, normalized so $P_{2017} = 1$.

Persistence Filter To reduce noise from temporary fluctuations, I zero out single-year spikes. Flagged exit must persist for at least two consecutive years:

$$\theta_{k,t} \leftarrow \theta_{k,t} \times \mathbf{1}\left[\theta^{\text{raw}}_{k,t-1} > 0 \text{ or } \theta^{\text{raw}}_{k,t+1} > 0\right].$$

B.5 Tariff Data Sources

U.S. tariffs on China (τ^d) are from Amiti, Redding, and Weinstein (2020) for monthly HS10 rates through 2017, aggregated to annual HS6 using 2017 import shares as weights. From 2018–2023, I use the Global Tariff Database from Teti (2025) and Rodríguez-Clare, Ulate, and Vasquez (2025), which provides bilateral HS6 tariffs. Hub country tariffs (τ^{hub}) are MFN rates from the same source.

C Empirical Results

Table C.1: Summary Statistics

	Pre		Post			
Variable	Mean	SD	Mean	SD	Post−Pre	N
$\Delta_{k,t}$	0.025	0.066	0.121	0.062	0.096	4246
Consumption	0.039	0.083	0.096	0.065	0.057	942
Capital	0.015	0.081	0.132	0.062	0.117	608
Intermediate	0.022	0.055	0.127	0.059	0.106	2696
$\theta_{a,\text{low}}$ (avoidance, cons.)	0.014	0.093	0.037	0.133	0.023	4246
Consumption	0.006	0.054	0.016	0.078	0.010	942
Capital	0.018	0.105	0.048	0.146	0.030	608
Intermediate	0.016	0.100	0.041	0.144	0.026	2696
$\theta_{a,\text{high}}$ (avoidance, lib.)	0.038	0.161	0.079	0.212	0.042	4246
Consumption	0.021	0.111	0.034	0.123	0.013	942
Capital	0.045	0.173	0.103	0.231	0.058	608
Intermediate	0.042	0.172	0.090	0.229	0.048	2696
Share($\theta_{a,\text{low}} > 0$)	0.098	0.297	0.248	0.432	0.150	4246
Share($\theta_{a,\text{high}} > 0$)	0.110	0.313	0.263	0.440	0.153	4246
$\theta_{p,\text{low}}$ (cessation, cons.)	0.000	0.000	0.061	0.238	0.061	4246
Consumption	0.000	0.000	0.047	0.211	0.047	942
Capital	0.000	0.000	0.039	0.195	0.039	608
Intermediate	0.000	0.000	0.070	0.255	0.070	2696
$\theta_{p,\text{high}}$ (cessation, lib.)	0.000	0.000	0.054	0.226	0.054	4246
Consumption	0.000	0.000	0.040	0.197	0.040	942
Capital	0.000	0.000	0.038	0.191	0.038	608
Intermediate	0.000	0.000	0.062	0.242	0.062	2696
Consumption share (2017)	0.203	0.385	0.203	0.385	0.000	4246
Capital share (2017)	0.149	0.348	0.149	0.348	0.000	4246
Intermediate share (2017)	0.648	0.463	0.648	0.463	0.000	4246

Table C.2: Transshipment Response to Tariffs

	Intensive Margin				Extensive Margin			
	θ_{low}		θ_{high}		$\mathbf{1}(\theta_{\text{low}} > 0)$		$\mathbf{1}(\theta_{\text{high}} > 0)$	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta_{k,t}$	0.263***		0.913***		2.02***		2.08***	
	(0.045)		(0.141)		(0.539)		(0.528)	
$\Delta_{k,t} \times$ Cap		0.251***		0.942***		1.92***		1.99***
		(0.062)		(0.221)		(0.662)		(0.632)
$\Delta_{k,t} \times$ Con		0.150***		0.401***		1.16		1.23
		(0.045)		(0.134)		(0.842)		(0.833)
$\Delta_{k,t} \times$ Int		0.480***		1.71***		3.71***		3.71***
		(0.090)		(0.199)		(0.453)		(0.474)
R^2	0.39	0.40	0.48	0.49	0.51	0.52	0.52	0.52
Observations	54,912	54,912	54,912	54,912	54,912	54,912	54,912	54,912
HS6 & Year×Use FE	✓	✓	✓	✓	✓	✓	✓	✓
Weighted	✓	✓	✓	✓	✓	✓	✓	✓

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the coefficients for regressions of (12). Each HS6–year observation is fractionally allocated across end-uses according to its UN-BEC shares. I estimate on the stacked HS6–year–end-use panel with HS6 and year × end-use fixed effects; weights are pre-period end-use share multiplied by the HS6 code's share of imports from China to the United States, and standard errors are clustered by HS6.

C.1 Pre-Trends

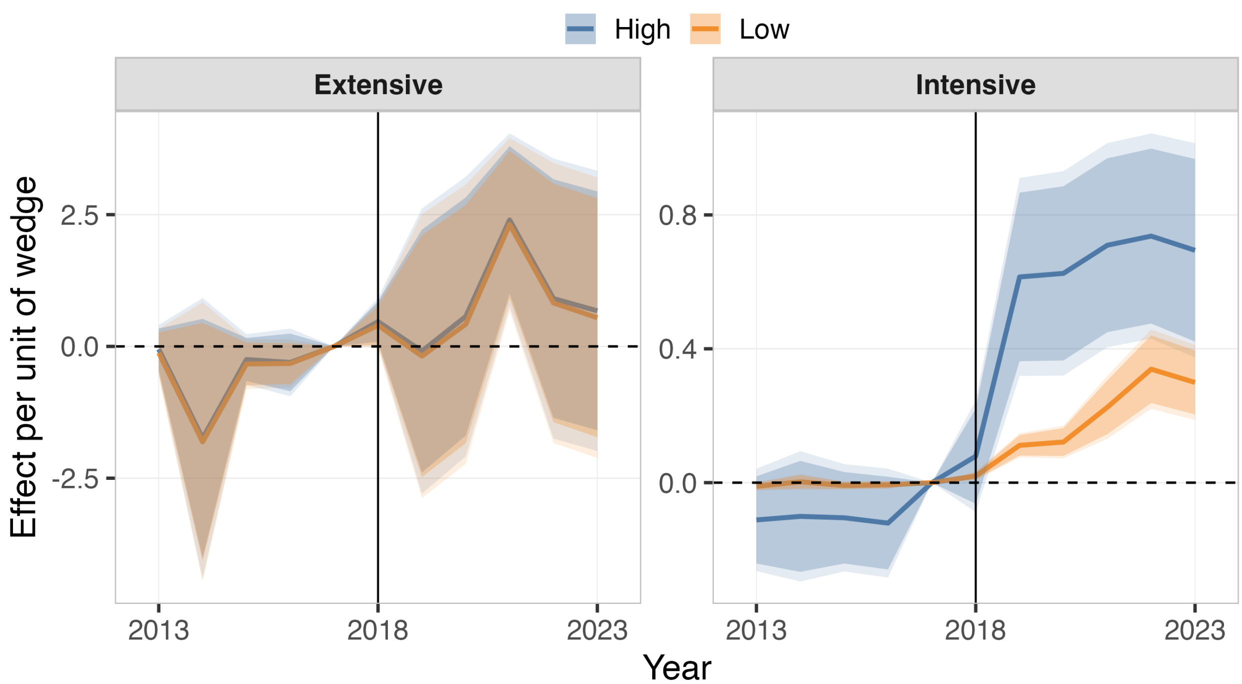
Figure 2 suggests that the tariffs caused transshipment to increase, but aggregate time series cannot establish causality: transshipment might have been rising for unrelated reasons, or the screens might mechanically flag any trade reallocation.

For each HS6 code k, I define exposure as the product's tariff wedge in 2019, $\Delta_{k,2019}$—the first year of full implementation.[14] The event-study specification is

$$\theta_{k,t} = \alpha_k + \delta_{t\times e(k)} + \sum_{r\neq -1} \beta_r \, \mathbf{1}\{\text{rel}_t = r\} \times \Delta_{k,2019} + \varepsilon_{k,t}, \tag{A.3}$$

where rel_t denotes years relative to 2018, α_k are HS6 fixed effects, and $\delta_{t\times e(k)}$ are year×end-use fixed effects. The outcome is $\theta_{k,t}$ on the intensive margin and $\mathbf{1}(\theta_{k,t} > 0)$ on the extensive margin. Observations are weighted by 2017 import shares and standard errors are clustered by HS6. Each β_r measures the difference in transshipment between high- and low-exposure products in year $2018 + r$, relative to 2017.

Figure C.1: Event Study: Transshipment Response to Tariff Exposure



Notes: Coefficients β_r from equation (A.3). Extensive margin (left) and intensive margin (right), with 90% and 95% confidence intervals. Conservative screen in orange; liberal screen in blue.

Figure C.1 plots the coefficients. Pre-trends are flat: from 2013 through 2017, intensive-margin coefficients are small and statistically indistinguishable from zero under both screens, and extensive-margin coefficients are noisy but centered near zero. Block-permutation tests confirm this for three of the four measures (Table C.3); the conservative intensive margin shows a small pre-period gap ($p = 0.005$). A direct pre-period interaction test yields small, insignifi-

14. Anchoring exposure at 2019 pins coefficients to the response per unit of the long-run wedge rather than time-varying contemporaneous exposure. Results are robust to alternative anchoring years (Supplemental Appendix Figure S1).

cant coefficients (Supplemental Appendix Table S5).[15] The post-2018 response is sharp on the intensive margin, with coefficients rising in 2019 and remaining elevated through 2023 under both screens. The extensive margin is less precisely estimated: post-period point estimates are positive, individually significant in 2021, and otherwise have confidence intervals that span zero.

Table C.3: Permutation Tests for Pre-Period Balance

Outcome	Observed Gap	P (left)	P (right)	P (two-sided)	P (Holm)
θ_{high}	0.006	0.964	0.036	0.072	0.216
θ_{low}	0.001	0.999	0.001	0.001	0.005
$\mathbf{1}\{\theta_{\text{high}} > 0\}$	-0.007	0.096	0.904	0.193	0.386
$\mathbf{1}\{\theta_{\text{low}} > 0\}$	0.001	0.132	0.868	0.263	0.386

Notes: Each row reports the observed weighted mean difference (treated minus placebo) in the specified outcome during the pre-period (2012–2017), along with permutation-based p-values. For each outcome, 5,000 block permutations were performed by reassigning treated HS6 codes within HS4 product groups while preserving the overall number of treated observations. For each permutation draw, the treated–placebo gap in the weighted mean outcome (weighted by 2017 China–US import shares) was recalculated. The empirical p-values are computed as the fraction of permuted gaps that are as or more extreme than the observed gap, with Holm–Bonferroni adjustments applied for multiple outcomes. All outcomes are defined at the HS6–year level and correspond to the conservative and liberal diversion measures in the main text, expressed as either semi-elasticity or binary indicators.

15. Main-specification coefficients are also robust to residualizing HS6-specific pre-trends (Supplemental Appendix Table S6) and to collapsing the panel to long differences (Supplemental Table S7).

C.2 Additional Results

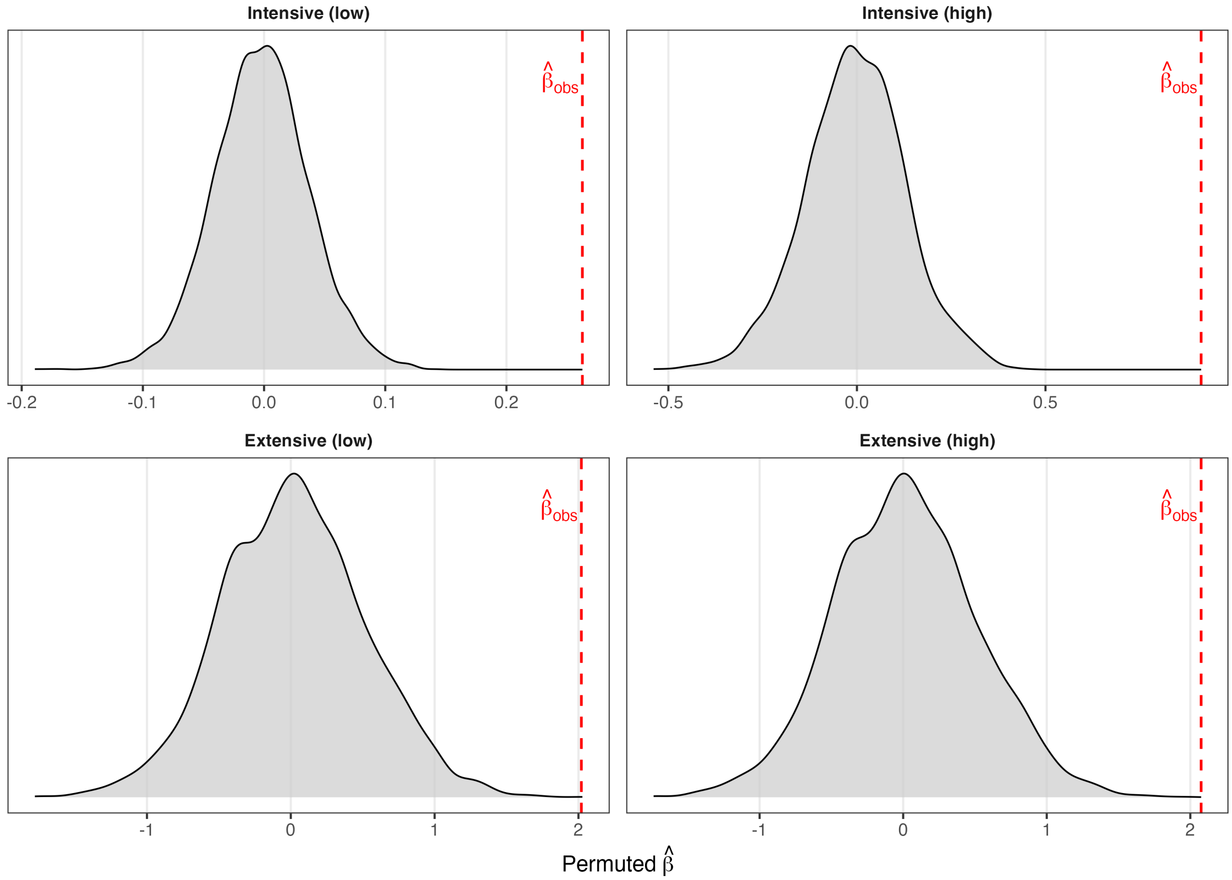
Table C.4: EU-27+UK and Canada Destination Placebo Results

EU-27+UK	θ_{low}		θ_{high}		$\mathbf{1}(\theta_{\text{low}} > 0)$		$\mathbf{1}(\theta_{\text{high}} > 0)$	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta_{k,t}$	0.016	0.018	0.033	0.034	0.073	0.095	0.047	0.106
	(0.011)	(0.015)	(0.022)	(0.026)	(0.149)	(0.101)	(0.158)	(0.104)
$\Delta_{k,t} \times \mathbf{1}(t \geq 2018)$		-0.002		-0.004		-0.024		-0.067
		(0.006)		(0.011)		(0.115)		(0.141)
Observations	54,564	54,564	54,564	54,564	54,564	54,564	54,564	54,564
R^2	0.31	0.31	0.32	0.32	0.36	0.36	0.37	0.37
Canada	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta_{k,t}$	0.051***	0.049*	0.061**	0.012	0.125	0.091	0.162	0.057
	(0.019)	(0.028)	(0.025)	(0.042)	(0.178)	(0.187)	(0.187)	(0.202)
$\Delta_{k,t} \times \mathbf{1}(t \geq 2018)$		0.003		0.057		0.039		0.121
		(0.017)		(0.037)		(0.315)		(0.318)
Observations	52,716	52,716	52,716	52,716	52,716	52,716	52,716	52,716
R^2	0.19	0.19	0.23	0.23	0.30	0.30	0.31	0.31
HS6 & Year×Use FE	✓	✓	✓	✓	✓	✓	✓	✓
Weighted	✓	✓	✓	✓	✓	✓	✓	✓

* p < 0.1, ** p < 0.05, *** p < 0.01

Notes: Entries report two-way fixed-effects regressions of placebo diversion measures on the U.S. tariff wedge, where the wedge is the same as in the main specification and is constructed from statutory U.S. HS6 tariffs and the tax-shield (see text). Outcomes are diversion measures computed with (i) the EU-27+UK treated as a single destination (top panel) and (ii) Canada as the destination (bottom panel). For each destination and year, diversion is built from HS6 flows using the same screening steps as in the main analysis but with the destination replaced by EU-27+UK or Canada. For comparability with the U.S. baseline, the exposure gate uses the U.S. HS6 tariff panel and diversion is capped within HS6 by the 2017 China→U.S. base so that $\sum_i \theta_{k,t} \leq 1$. Columns (1)–(2) and (5)–(6) use intensive-margin outcomes θ_{low} and θ_{high}; columns (3)–(4) and (7)–(8) use indicators $\mathbf{1}\{\theta_{\text{low}} > 0\}$ and $\mathbf{1}\{\theta_{\text{high}} > 0\}$. All specifications include HS6 and year×use fixed effects with standard errors clustered by HS6.

Figure C.2: Randomization Inference: Null Distributions of $\widehat{\beta}$ under Permuted Wedge Assignment



Notes: Each of 5,000 permutations randomly reassigns wedge trajectories across HS6 codes: each product k inherits the full time series of wedge values $\{\Delta_{k',t}\}_t$ from a randomly drawn donor product k', preserving within-product wedge dynamics but randomizing which product faces which trajectory. For each permutation I re-estimate equation (12) with HS6 and year×end-use fixed effects, 2017 China→US import-share weights, and HS6 clustering. Gray densities plot the resulting null distributions of $\widehat{\beta}$ across permutations. Red dashed lines mark the observed $\widehat{\beta}$ in the intact data. Top panels: intensive margin with outcomes θ_{low} (left) and θ_{high} (right). Bottom panels: extensive margin with outcomes $\mathbf{1}(\theta_{\text{low}} > 0)$ and $\mathbf{1}(\theta_{\text{high}} > 0)$. The observed $\widehat{\beta}$ falls outside the support of the simulated null in all four specifications.

Table C.5: Heterogeneous Fixed Costs Along the Intensive Margin

	θ_{low}			θ_{high}		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta_{k,t}$	0.252***	0.253***	0.196***	0.845***	0.843***	0.641***
	(0.044)	(0.043)	(0.035)	(0.140)	(0.139)	(0.105)
$\Delta_{k,t}$ × Pre-2017 Hub Intensity	0.065			0.398**		
	(0.053)			(0.163)		
$\Delta_{k,t}$ × Bulkiness		-0.044			-0.328**	
		(0.030)			(0.163)	
$\Delta_{k,t}$ × Electronics			0.088***			0.355***
			(0.028)			(0.090)
R^2	0.40	0.39	0.40	0.49	0.49	0.50
Observations	54,912	54,912	54,912	54,912	54,912	54,912
Weighted	✓	✓	✓	✓	✓	✓
HS6 & Year×Use FE	✓	✓	✓	✓	✓	✓

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Each column reports regressions of $\theta_{k,t}$ on the effective wedge and its interaction with a pre-2017 product characteristic. Hub intensity is the standardized share of China's pre-2017 exports of product k routed through third-country hubs. "Bulkiness" is kilograms per U.S. dollar of import value. "Electronics" is an indicator for HS2 codes 84, 85, or 90. HS6 and year×use fixed effects are included; observations are weighted by 2017 China→US exposure multiplied by pre-period end-use share; standard errors are clustered by HS6. Coefficients are per unit of the wedge (log gross factor). The interaction terms measure how the wedge effect varies across products with different pre-existing hub exposure, bulkiness, or electronics content.

Table C.6: Heterogeneous Fixed Costs Along the Extensive Margin

	$\mathbf{1}(\theta_{\text{low}} > 0)$			$\mathbf{1}(\theta_{\text{high}} > 0)$		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta_{k,t}$	1.95***	2.01***	1.76***	1.99***	2.04***	1.83***
	(0.540)	(0.528)	(0.549)	(0.530)	(0.518)	(0.539)
$\Delta_{k,t} \times$ Pre-2017 Hub Intensity	0.426			0.493		
	(0.412)			(0.405)		
$\Delta_{k,t} \times$ Bulkiness		-0.064			-0.188	
		(0.287)			(0.296)	
$\Delta_{k,t} \times$ Electronics			0.341			0.317
			(0.239)			(0.238)
R^2	0.51	0.51	0.52	0.52	0.52	0.52
Observations	54,912	54,912	54,912	54,912	54,912	54,912
Weighted	✓	✓	✓	✓	✓	✓
HS6 & Year×Use FE	✓	✓	✓	✓	✓	✓

* p < 0.1, ** p < 0.05, *** p < 0.01

Notes: The procedure is the same as in Table C.5, except we now have the extensive margin as the dependent variable.

Table C.7: Exit Intensity and Direct China–US Trade

	$\Delta^{\text{trend}}(\text{CN} \rightarrow \text{US})_{k,t}$			
	Quantities		Prices	
	(1)	(2)	(3)	(4)
θ_{low}	-1.07***		-0.983**	
	(0.297)		(0.394)	
θ_{high}		-0.652***		-0.412***
		(0.128)		(0.129)
R^2	0.46	0.47	0.34	0.34
Observations	53,485	53,485	53,134	53,134
HS6 FE	✓	✓	✓	✓
Year×Use FE	✓	✓		
Year FE			✓	✓
Weighted	✓	✓	✓	✓

* p < 0.1, ** p < 0.05, *** p < 0.01

Notes: Regressions of $\Delta^{\text{trend}}_{\text{CN}\rightarrow\text{US}}$—the deviation of HS6-level direct China→US imports from their 2012–2017 linear trend—on exit intensity. Columns 1–2: log import values, with HS6 and year×end-use fixed effects. Columns 3–4: China FOB unit values, with HS6 and year fixed effects. All specifications are weighted by pre-period China→US import shares; standard errors are clustered by HS6.

Table C.8: Numerator (Flagged Value) Regressions

	$\ln(FV_{low})$	$\ln(FV_{high})$	$\ln(FV_{low})$	$\ln(FV_{high})$
	(1)	(2)	(3)	(4)
$\Delta_{k,t}$	16.5***	20.3***	20.8***	24.9***
	(5.51)	(5.74)	(6.84)	(7.30)
$\Delta_{k,t}$ × China Mkt Share	7.11**	8.07**		
	(3.47)	(3.90)		
$\Delta_{k,t}$ × HHI			-0.657	-1.85
			(4.29)	(4.87)
R^2	0.54	0.53	0.54	0.52
Observations	54,912	54,912	54,912	54,912
HS6 & Year×Use FE	✓	✓	✓	✓
Weighted	✓	✓	✓	✓

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Regressions of the log of the HS6 flagged value (the numerator of θ) on the wedge and its interaction with a pre-tariff HS6 characteristic. Columns (1)–(2) interact with China market share; columns (3)–(4) interact with the pre-tariff exporter HHI. This isolates the numerator response and removes the mechanical dependence on the denominator (2017 China→US exposure). All regressions use HS6 and year×end-use fixed effects, are weighted by 2017 China→US exposure times pre-period end-use share, and cluster standard errors by HS6.

Table C.9: Market-Share Selection: Extensive Margin

	Full Sample		Entry-Only	
	$\mathbf{1}(\theta_{\text{low}} > 0)$	$\mathbf{1}(\theta_{\text{high}} > 0)$	$\mathbf{1}(\theta_{\text{low}} > 0)$	$\mathbf{1}(\theta_{\text{high}} > 0)$
	(1)	(2)	(3)	(4)
$\Delta_{k,t}$	1.66^{***}	1.73^{***}	2.17^{***}	2.26^{***}
	(0.459)	(0.449)	(0.601)	(0.590)
$\Delta_{k,t}$ × Pre-Tariff China Mkt Share	0.588^{**}	0.563^{**}	0.810^{***}	0.805^{***}
	(0.267)	(0.264)	(0.304)	(0.296)
R^2	0.52	0.52	0.64	0.64
Observations	54,912	54,912	34,092	32,484
HS6 & Year×Use FE	✓	✓	✓	✓
Weighted	✓	✓	✓	✓

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Extensive-margin regressions of diversion indicators on the tariff wedge and its interaction with the standardized pre-tariff China import market share at the HS6 level. Columns (1)–(2) use all HS6 codes; columns (3)–(4) restrict to the entry subsample (HS6 codes with zero pre-period θ). All specifications include HS6 and year×end-use fixed effects, are weighted by 2017 China→US exposure times pre-period end-use share, and cluster standard errors by HS6.

Table C.10: Market-Share Selection: Intensive Margin

	Full Sample		Entry-Only	
	θ_{low}	θ_{high}	θ_{low}	θ_{high}
	(1)	(2)	(3)	(4)
$\Delta_{k,t}$	0.324***	1.08***	0.321***	1.22***
	(0.067)	(0.197)	(0.086)	(0.295)
$\Delta_{k,t}$ × Pre-Tariff China Mkt Share	-0.100***	-0.275***	-0.096***	-0.357***
	(0.029)	(0.094)	(0.034)	(0.128)
R^2	0.40	0.49	0.43	0.55
Observations	54,912	54,912	34,092	32,484
HS6 & Year×Use FE	✓	✓	✓	✓
Weighted	✓	✓	✓	✓

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Intensive-margin analogue to Table C.9. The outcome is θ rather than $\mathbf{1}(\theta > 0)$. All other specifications, weights, fixed effects, and clustering are identical. The intensive-margin coefficient should be interpreted with the denominator caveat that θ is bounded above by 2017 China→US exposure.

Table C.11: Price-Distribution Shifts Across Quantiles

	P10	P25	P50	P75	P90
	(1)	(2)	(3)	(4)	(5)
θ_{high}	-0.157***	-0.076*	-0.006	0.009	0.046
	(0.059)	(0.044)	(0.052)	(0.057)	(0.103)
R^2	0.08	0.07	0.07	0.09	0.09
Observations	50,288	50,288	50,288	50,288	50,288
HS6 & Year×Use FE	✓	✓	✓	✓	✓
Weighted	✓	✓	✓	✓	✓

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Each column regresses the year-on-year change in the within-HS6 log unit-value quantile (P10 through P90, computed across exporters and value-weighted) on θ_{high}. All specifications include HS6 and year×end-use fixed effects, are weighted by 2017 China→US exposure times pre-period end-use share, and cluster standard errors by HS6. The coefficient traces how diversion shifts the shape of the surviving direct-shipment price distribution.

Table C.12: Exit Response by Rauch Differentiation

	Intensive		Extensive	
	θ_{low}	θ_{high}	$\mathbb{1}(\theta_{\text{low}} > 0)$	$\mathbb{1}(\theta_{\text{high}} > 0)$
Panel A: Differentiated goods				
$\Delta_{k,t}$	0.288***	1.002***	2.118***	2.117***
	(0.041)	(0.136)	(0.591)	(0.591)
N	36,324	36,324	36,324	36,324
R^2	0.408	0.491	0.502	0.503
Panel B: Non-differentiated goods				
$\Delta_{k,t}$	0.117	0.354*	1.256**	1.602***
	(0.091)	(0.194)	(0.616)	(0.607)
N	18,588	18,588	18,588	18,588
R^2	0.355	0.473	0.622	0.631
HS6 & Year×Use FE	✓	✓	✓	✓

* p < 0.1, ** p < 0.05, *** p < 0.01

Notes: Baseline wedge regression $\theta = \beta\,\Delta_{k,t} + \text{FE} + \varepsilon$ estimated separately on differentiated (Panel A) and non-differentiated (Panel B) HS6 codes under the Rauch (1999) classification. Intensive-margin outcomes are θ_{low} and θ_{high}. Extensive-margin outcomes are indicators for whether $\theta > 0$ under each screen. All specifications include HS6 and year×end-use fixed effects, weight by 2017 China-to-US exposure times pre-period end-use share, and cluster standard errors at the HS6 level.

Supplemental Appendix

S.1 Supplemental Theoretical Results

S.1.0.1 Cross-Side Equivalence

The baseline model assigns channel-choice and price decisions to the exporter. In practice, U.S. importers and trade intermediaries also have incentives to arbitrage across direct and rerouted channels. The following lemma shows that the channel partition characterized in Proposition 4 is invariant to which side of the transaction bears the fixed costs, provided the parties bargain efficiently.

Lemma 2 (Cross-side equivalence)**.** *Consider an exporter with marginal cost m and a U.S. importer who buys the variety and resells competitively to U.S. consumers at the delivered price $(1+\tau^r)p$. Suppose the parties engage in efficient bargaining over channel choice $r \in \{d, a\}$, the operating decision, and the price p, with any protocol that selects an outcome on the joint-surplus frontier. Define joint surplus by*

$$J(m, 1+\tau^d) \equiv V(m, 1+\tau^d) - F_d, \qquad J(m, 1+\tau^a) \equiv V(m, 1+\tau^a) - F_d - F_a.$$

The cutoffs $\widehat{m}$ and $\bar{m}$ of Proposition 4 satisfy

$$\Delta V(\widehat{m}) = F_a, \qquad V(\bar{m}, 1+\tau^d) = F_d,$$

and depend only on $J(\cdot)$. The partition is therefore invariant to how F_a and F_d are allocated between exporter and importer.

Proof. Because the importer resells competitively, importer profit per transaction is zero and joint variable surplus from operating in channel r at price p equals exporter variable profit $(p-m)D((1+\tau^r)p \mid \mathbf{P})$. Fixed costs F_a and F_d enter joint surplus identically regardless of which party writes the check, since either party paying out of pocket reduces the surplus available to divide between them. Maximizing variable surplus over p in each channel yields $V(m, 1+\tau^r)$, and subtracting the relevant fixed costs gives $J(m, 1+\tau^r)$ as defined.

Under efficient bargaining, the parties select $(r, p, \text{operate})$ to maximize J. The decision to reroute is $J(m, 1+\tau^a) \geq J(m, 1+\tau^d)$. Substituting,

$$V(m, 1+\tau^a) - F_d - F_a \geq V(m, 1+\tau^d) - F_d \iff \Delta V(m) \geq F_a,$$

yielding the avoidance cutoff $\Delta V(\widehat{m}) = F_a$. The decision to operate at all is $\max_r J(m, 1+\tau^r) \geq 0$, which under Assumption 2 reduces to $V(m, 1+\tau^d) \geq F_d$, yielding the participation cutoff $V(\bar{m}, 1+\tau^d) = F_d$. Both conditions reference only $J(\cdot)$ and are invariant to the cost split.

The bargaining protocol determines how the realized surplus J is divided between exporter and importer through transfers but does not affect the channel chosen or the cutoffs at which channel choice flips. $\square$

The lemma rests on efficient bargaining, which selects an outcome on the joint-surplus frontier. Under inefficient protocols—double marginalization, hold-up, or take-it-or-leave-it offers that exclude positive-surplus trades—the partition may shift in level, but the existence of two opposing exit margins is preserved.

S.1.1 Heterogeneous Fixed Costs

This appendix allows both fixed costs — the rerouting setup cost F_a and the direct-channel operating cost F_d — to vary across firms or products. Both cutoffs in Proposition 4 then become firm-specific. The partition holds pointwise; the dominance condition of Corollary 2 continues to govern the sign of the survivor-cohort gap; the bound of Proposition 2 holds at the aggregate.

Corollary 4 (Heterogeneous fixed costs)**.** *Let Assumptions **D1–D5** hold, and let $(F_{a,i}, F_{d,i})$ be drawn from a continuous joint distribution G_F on $[F_{a,\min}, F_{a,\max}] \times [F_{d,\min}, F_{d,\max}]$, independent of m_i, with $F_{a,\min} \geq \Delta V(\bar{m}(F_{d,\max}; \tau^d))$ so that Assumption 2 holds for every realization.*[16]

(i) ***Firm-specific cutoffs and partition.*** *For each realization (F_a, F_d), Proposition 4 delivers cutoffs $\widehat{m}(F_a; \tau^d, \tau^a)$ solving $\Delta V(\widehat{m}) = F_a$ and $\bar{m}(F_d; \tau^d)$ solving $V(\bar{m}, 1+\tau^d) = F_d$, with $\widehat{m} \leq \bar{m}$. Firm i reroutes if $m_i < \widehat{m}_i \equiv \widehat{m}(F_{a,i}; \cdot)$, ships directly if $\widehat{m}_i \leq m_i \leq \bar{m}_i \equiv \bar{m}(F_{d,i}; \cdot)$, and exits otherwise. The comparative statics of Proposition 4 hold pointwise; additionally $\partial\widehat{m}/\partial F_a < 0$ and $\partial\bar{m}/\partial F_d < 0$.*

(ii) ***Selection on each margin.*** *Under Kimball curvature, direct-channel passthrough $\beta(m)$ is increasing in m. Survival in the direct sample $\sigma_i = \mathbf{1}\{\widehat{m}_i \leq m_i \leq \bar{m}_i\}$ is a window indicator: it jumps from 0 to 1 at $\widehat{m}_i$ (the avoidance margin removes low-β firms) and from 1 to 0 at $\bar{m}_i$ (the participation margin removes high-β firms). The two margins push the survivor-cohort gap in opposite directions, exactly as in Corollary 2.*

(iii) ***Aggregate covariance and bound.*** *Let $\theta_a \equiv \Pr(m_i < \widehat{m}_i)$ and $\theta_p \equiv \Pr(m_i > \bar{m}_i)$, with $\theta = \theta_a + \theta_p$. Under avoidance dominance (Corollary 2), $\mathrm{Cov}_\omega(\beta_i, \sigma_i) \geq 0$ and the case-(i)*

16. Since $\bar{m}$ is decreasing in F_d and ΔV is decreasing in m, $\Delta V(\bar{m}(F_d; \tau^d))$ is increasing in F_d; the binding case is $F_d = F_{d,\max}$.

bound of Proposition 2 applies:

$$\beta_{cohort} \in \big[(1-\theta)\,\beta_{survivor} + \theta\,\beta_{\min},\ \beta_{survivor}\big].$$

Under participation dominance, the case-(ii) bound applies instead. The dominance regime is determined by the joint distribution of (F_a, F_d) together with G, not by either fixed cost in isolation.

(iv) ***Cross-category heterogeneity.*** *If (F_a, F_d) is deterministic by end-use category k, all conclusions hold within k. Lower $F_{a,k}$ raises $\widehat{m}_k$ and increases $\theta_{a,k}$; lower $F_{d,k}$ raises $\bar{m}_k$ and decreases $\theta_{p,k}$. Cross-category variation in F_a provides a structural rationale for the larger avoidance shares observed in capital and intermediate goods (Section 5); the empirical finding $\theta_{p,k} \approx 0$ across categories (Section 5.1) is consistent with $F_{d,k}$ being uniformly small, so that direct-channel variable profit clears the operating cost across the relevant range of m and few firms hit the participation margin.*

Proof. For each realization (F_a, F_d), Proposition 4 delivers existence, uniqueness, and the sign pattern of the cutoffs. The IFT gives $\partial\widehat{m}/\partial F_a = 1/\Delta V_m < 0$ and $\partial\bar{m}/\partial F_d = 1/V_m < 0$ since both denominators are negative. For (ii), the indicator σ_i is a window in m_i for each realization of $(F_{a,i}, F_{d,i})$; the contributions of the two cut points to $\mathrm{Cov}_\omega(\beta_i, \sigma_i \mid F_a, F_d)$ enter with opposite signs, with avoidance contributing positively and participation contributing negatively, by the same logic as Corollary 2. By the law of total covariance, $\mathrm{Cov}_\omega(\beta_i, \sigma_i) = \mathbb{E}_{F_a,F_d}[\mathrm{Cov}_\omega(\beta_i, \sigma_i \mid F_a, F_d)] + \mathrm{Cov}_{F_a,F_d}(\mathbb{E}[\beta_i \mid F_a, F_d], \mathbb{E}[\sigma_i \mid F_a, F_d])$; under independence the second term vanishes since $\mathbb{E}[\beta_i \mid F_a, F_d] = \mathbb{E}[\beta_i]$ is constant in (F_a, F_d), so the aggregate covariance is the expectation of conditional covariances and inherits their sign whenever they are uniformly signed. The bound from Proposition 2 then applies. Category-specific statements are immediate by conditioning on k. □

Remarks. Independence of (F_a, F_d) from m is sufficient, not necessary. The results extend whenever the joint distribution preserves the dominance ordering: a positive association between F_a and m would strengthen avoidance dominance (high-cost firms face higher F_a and are even less likely to reroute); symmetrically, a positive association between F_d and m would strengthen participation dominance (high-cost firms face higher F_d and are even more likely to shut down). A sufficiently strong reversal of either association could in principle flip the dominant margin.

S.1.2 Variable Rerouting Costs

Section 3.1 treats per-unit rerouting costs as isomorphic to a higher effective tariff rate. This appendix verifies the isomorphism formally. Variable rerouting costs affect only the avoidance margin—the participation cutoff $\bar{m}$ is unchanged because per-unit rerouting costs do not enter direct-channel profit.

Suppose rerouting raises the exporter's marginal cost from m to $m + \Delta c$ in addition to facing the lower effective tariff τ^a. Optimized variable profit in the rerouting channel becomes $V(m + \Delta c, 1 + \tau^a)$. The avoidance cutoff $\widehat{m}_\Delta$ solves

$$V(\widehat{m}_\Delta + \Delta c, 1 + \tau^a) - F_a \ = \ V(\widehat{m}_\Delta, 1 + \tau^d).$$

The participation cutoff is $\bar{m}$, defined as in Proposition 4 by $V(\bar{m}, 1 + \tau^d) = F_d$. The corollary below covers the small-Δc regime in which the cost penalty of rerouting does not exceed the tariff benefit at any cost level; for very large Δc the rerouting channel can become uniformly dominated by direct shipping and the avoidance region collapses.

Corollary 5 (Selection under variable rerouting costs)**.** *Under* ***D1*****–*****D4****, suppose* Δc *satisfies* $q^a(m+\Delta c) > q^d(m)$ *over* $[\underline{m}, \bar{m}]$*, and that* $V(\underline{m} + \Delta c, 1 + \tau^a) - F_a > V(\underline{m}, 1 + \tau^d)$ *so the lowest-cost exporter strictly prefers rerouting. Then there exists a unique avoidance cutoff* $\widehat{m}_\Delta(\tau^d, \tau^a, F_a, \Delta c) \in [\underline{m}, \bar{m}]$ *such that*

$$m < \widehat{m}_\Delta \Rightarrow \textit{Reroute}, \quad \widehat{m}_\Delta \leq m \leq \bar{m} \Rightarrow \textit{Direct}, \quad m > \bar{m} \Rightarrow \textit{Exit}.$$

The avoidance cutoff satisfies $\partial\widehat{m}_\Delta/\partial\tau^d > 0$*,* $\partial\widehat{m}_\Delta/\partial\tau^a < 0$*,* $\partial\widehat{m}_\Delta/\partial F_a < 0$*, and* $\partial\widehat{m}_\Delta/\partial\Delta c < 0$*. The participation cutoff* $\bar{m}$ *is unaffected by* Δc*.*

Proof. Define $\tilde{H}(m; \Delta c) \equiv V(m + \Delta c, 1 + \tau^a) - F_a - V(m, 1 + \tau^d)$. By the envelope theorem, $\partial\tilde{H}/\partial m = -q^a(m + \Delta c) + q^d(m)$. Under **D1**–**D4** and $\tau^a < \tau^d$, the rerouting channel has higher demand at any common marginal cost. For Δc small enough that $q^a(m + \Delta c) > q^d(m)$ over the relevant range, $\partial\tilde{H}/\partial m < 0$. The premise $\tilde{H}(\underline{m}; \Delta c) > 0$ combined with continuity implies a unique cutoff $\widehat{m}_\Delta$ at which $\tilde{H} = 0$. The comparative statics follow from the IFT applied at the cutoff. The participation cutoff $\bar{m}$ depends only on $V(\cdot, 1 + \tau^d)$ and F_d, which are unaffected by Δc. □

The corollary establishes that the variable-cost extension preserves the three-region partition of Proposition 4, with $\widehat{m}_\Delta$ shifted leftward relative to the $\Delta c = 0$ benchmark while $\bar{m}$ remains fixed. The structural input that Section 2 requires—the sign of $\mathrm{Cov}_\omega(\beta_i, \sigma_i)$—is unchanged.

S.1.3 Heterogeneous Avoidance Costs and Tax Deductibility

Section 3.1 notes that per-unit avoidance costs $\Delta c \geq 0$ are isomorphic to a higher effective tax rate in the avoidance channel. This appendix develops the cross-product implications and shows that tax deductibility of imported inputs is a natural source of heterogeneous effective avoidance costs. Within each product, only the avoidance cutoff $\widehat{m}_k$ varies with deductibility; the participation cutoff $\bar{m}_k$ depends on τ_k^d and F_d, not on τ_k^a, z_k, or τ^c.

Heterogeneous effective wedges. Let the effective avoidance wedge τ_k^a vary across products k. Within each product, τ_k^a is common across firms, so Proposition 4 applies directly: there exists a product-specific avoidance cutoff $\widehat{m}_k = \widehat{m}(\tau_k^d, \tau_k^a, F_a)$ solving $\Delta V_k(\widehat{m}_k) = F_a$ and a product-specific participation cutoff $\bar{m}_k = \bar{m}(\tau_k^d, F_d)$. Products with higher effective avoidance costs have smaller avoidance shares $\theta_{a,k} = G(\widehat{m}_k)$. Monotone selection (under avoidance dominance), the survivor–cohort bias, and the case-(i) incidence bound all hold within each product:

$$\beta_{\text{cohort},k} \in \left[(1-\theta_{a,k})\, \beta_{\text{survivor},k} + \theta_{a,k}\, \beta_{\min},\ \beta_{\text{survivor},k}\right].$$

The relevant policy variable is the effective wedge $\Delta_k \equiv \ln \frac{1+\tau_k^d}{1+\tau_k^a}$: products with a larger wedge have more avoidance.

Deductibility as a source of heterogeneous τ^a. When domestic firms deduct tariff-inclusive import costs from taxable income, the after-tax cost of importing through the direct channel falls. This compresses the gap between channels, reducing the incentive to reroute. Suppose the importer faces corporate tax rate $\tau^c \in [0, 1)$ and deducts a fraction $z_k \in [0, 1]$ of the tariff-inclusive cost, where z_k is the net present value of deductions for product k. For intermediates expensed immediately, $z_k = 1$; for capital goods depreciated over time, $z_k < 1$; for consumption goods purchased by non-deducting households, $z_k = 0$. The effective avoidance wedge under deductibility is

$$\tau_k^a = z_k\, \tau^c\, \tau_k^d,$$

yielding

$$\Delta_k = \ln \frac{1+\tau_k^d}{1+z_k\, \tau^c\, \tau_k^d}.$$

Three comparative statics follow immediately:

(a) $\partial \Delta_k / \partial \tau^c < 0$, so a higher corporate tax rate compresses the wedge, discouraging avoidance.

(b) $\partial \Delta_k / \partial z_k < 0$, so greater deductibility compresses the wedge.

(c) $\partial\Delta_k/\partial\tau_k^d > 0$, so a higher statutory tariff widens the wedge, but the effect is attenuated by deductibility ($\partial^2\Delta_k/\partial\tau_k^d\,\partial\tau^c < 0$).

S.1.3.1 The 2017 Tax Cuts and Jobs Act

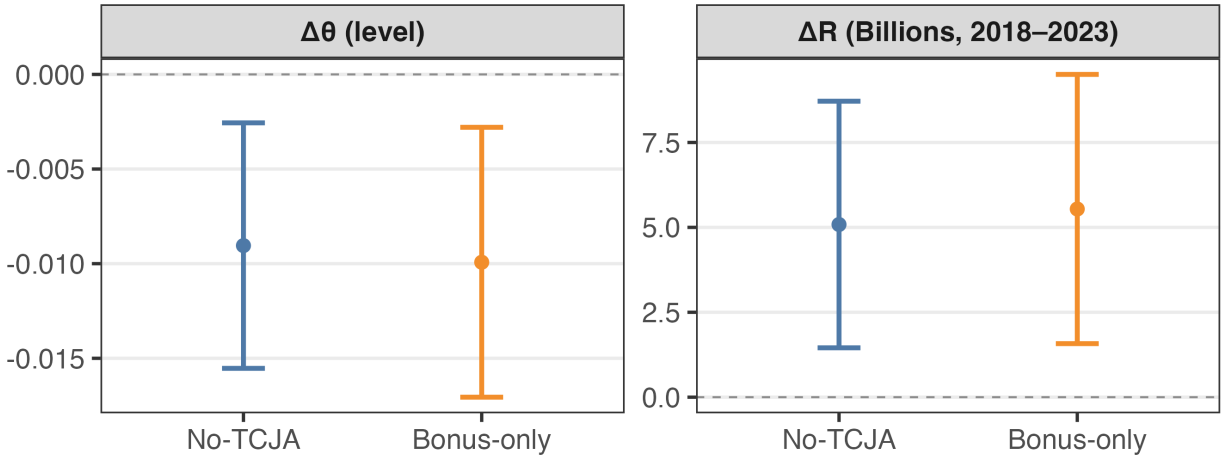
The exit channel documented above creates demand for policy levers that curb avoidance. The traditional tools are explicit enforcement: audits, penalties, and customs verification (Allingham and Sandmo 1972; Slemrod and Yitzhaki 2002; Slemrod 2019). A growing literature studies how the design of other tax instruments shapes the incentive to exit a tax base—most directly, Blandhol (2025) shows that out-migration taxes meaningfully reduce capital flight following a Norwegian wealth-tax reform. This paper identifies an analogous channel inside the U.S. corporate tax code: deductibility of tariff-inclusive import costs acts as implicit enforcement on rerouting.

U.S. importers deduct the tariff-inclusive cost of imports from taxable income, while rerouting forfeits this deduction. A higher corporate rate τ^c therefore raises the implicit cost of avoidance and compresses the wedge $\Delta_k = \log\left((1+\tau_k^d)/(1+\tau_k^a)\right)$, where $\tau_k^a = z_k\tau^c\tau_k^d$ (Appendix S.1.3). Compression is proportional to the statutory tariff. The cross-end-use ordering of exit is therefore set by statutory rates—intermediates highest, consumption lowest—and deductibility shifts wedge levels uniformly without reordering them.[17] The 2017 Tax Cuts and Jobs Act cut τ^c from 35% to 21% and accelerated depreciation for equipment just before the 2018 tariffs. The corporate rate cut widened Δ_k for most imports; the depreciation acceleration partially offset this for capital goods.

To separate the two channels, I compute two counterfactuals. *No-TCJA* holds τ^c at 35% and applies the pre-TCJA bonus-depreciation schedule. *Bonus-only* holds τ^c at 35% but retains 100% bonus depreciation. Figure S1 reports the change in aggregate exit share and cumulative tariff revenue over 2018–2023 under each counterfactual relative to the observed path, with Imbens–Manski bands reflecting uncertainty in ψ. Under no-TCJA, rerouting would have been 0.3 to 1.6 percentage points lower and cumulative revenue \$2 to \$9 billion higher. The bonus-only counterfactual is much smaller because it affects only capital-goods imports, which were already subject to substantial bonus depreciation pre-TCJA.

17. Post-TCJA weighted-mean effective wedges average 7.5 percentage points for consumption, 8.3 for capital, and 13.3 for intermediates, a gap mirroring Section 301 tariffs of roughly 20% on intermediate imports versus 10% on consumption. The cross-end-use ordering is robust to alternative deductibility assumptions; the statutory differential dominates.

Figure S1: Counterfactual Impact of the 2017 Tax Cuts and Jobs Act on Exit and Tariff Revenue



Notes: Change in aggregate exit share $\Delta\theta$ and cumulative tariff revenue ΔR over 2018–2023 under two counterfactuals relative to observed outcomes. "No-TCJA" holds $\tau^c = 35\%$ and applies the pre-TCJA bonus-depreciation schedule. "Bonus-only" holds $\tau^c = 35\%$ but retains 100% bonus depreciation. Vertical bars are 95% Imbens–Manski joint confidence sets.

S.2 Supplemental Empirical Results

Table S1: Transshipment Response to Tariffs (Clustered by HS4 × year)

	Intensive Margin				Extensive Margin			
	θ_{low}		θ_{high}		$\mathbf{1}(\theta_{\text{low}} > 0)$		$\mathbf{1}(\theta_{\text{high}} > 0)$	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta_{k,t}$	0.263***		0.913***		2.02***		2.08***	
	(0.064)		(0.152)		(0.609)		(0.589)	
$\Delta_{k,t}$ × Cap		0.251***		0.942***		1.92**		1.99**
		(0.076)		(0.235)		(0.713)		(0.676)
$\Delta_{k,t}$ × Con		0.150**		0.401**		1.16		1.23
		(0.050)		(0.137)		(0.832)		(0.815)
$\Delta_{k,t}$ × Int		0.480***		1.71***		3.71***		3.71***
		(0.122)		(0.198)		(0.512)		(0.532)
R^2	0.39	0.40	0.48	0.49	0.51	0.52	0.52	0.52
Observations	54,912	54,912	54,912	54,912	54,912	54,912	54,912	54,912
HS6 & Year×Use FE	✓	✓	✓	✓	✓	✓	✓	✓
Weighted	✓	✓	✓	✓	✓	✓	✓	✓

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the coefficients for regressions of (12). Each HS6–year observation is fractionally allocated across end-uses according to its UN-BEC shares. I estimate on the stacked HS6–year–end-use panel with HS6 and year × end-use fixed effects; weights are pre-period end-use share multiplied by the HS6 code's share of imports from China to the United States, and standard errors are clustered by HS4 and year.

Table S2: Transshipment Response to Tariffs (unweighted)

	Intensive Margin				Extensive Margin			
	θ_{low}		θ_{high}		$\mathbf{1}(\theta_{\text{low}} > 0)$		$\mathbf{1}(\theta_{\text{high}} > 0)$	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta_{k,t}$	0.108^{***}		0.216^{***}		0.523^{***}		0.529^{***}	
	(0.016)		(0.033)		(0.066)		(0.067)	
$\Delta_{k,t} \times$ Cap		0.106^{***}		0.155^{*}		0.336^{***}		0.295^{**}
		(0.036)		(0.086)		(0.126)		(0.120)
$\Delta_{k,t} \times$ Con		0.043^{***}		0.095^{***}		0.356^{***}		0.347^{***}
		(0.015)		(0.029)		(0.091)		(0.092)
$\Delta_{k,t} \times$ Int		0.149^{***}		0.317^{***}		0.707^{***}		0.742^{***}
		(0.028)		(0.052)		(0.106)		(0.112)
R^2	0.23	0.23	0.27	0.27	0.32	0.32	0.32	0.32
Observations	54,912	54,912	54,912	54,912	54,912	54,912	54,912	54,912
HS6 & Year×Use FE	✓	✓	✓	✓	✓	✓	✓	✓
Weighted	✓	✓	✓	✓	✓	✓	✓	✓

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the coefficients for regressions of (12). Each HS6–year observation is fractionally allocated across end-uses according to its UN-BEC shares. I estimate on the stacked HS6–year–end-use panel with HS6 and year × end-use fixed effects; weights are pre-period end-use share (since I are stacking by end-use, this is the same as unweighted), and standard errors are clustered by HS6.

Table S3: Transshipment Response to Tariffs (No Deductibility)

	θ_{low}		θ_{high}	
	(1)	(2)	(3)	(4)
$(1+\tau^d_{k,t})$	0.123***		0.449***	
	(0.038)		(0.126)	
$(1+\tau^d_{k,t}) \times$ Cap		0.108**		0.411**
		(0.053)		(0.183)
$(1+\tau^d_{k,t}) \times$ Con		0.091***		0.231***
		(0.029)		(0.083)
$(1+\tau^d_{k,t}) \times$ Int		0.244***		1.04***
		(0.050)		(0.147)
R^2	0.39	0.39	0.47	0.48
Observations	54,912	54,912	54,912	54,912
HS6 & Year×Use FE	✓	✓	✓	✓
Weighted	✓	✓	✓	✓

* p < 0.1, ** p < 0.05, *** p < 0.01

Notes: This table repeats the intensive margin exercise from Table C.2, but uses the tariff wedge gross of the deductibility penalty as a regressor. Each HS6–year observation is fractionally allocated across end-uses according to its UN-BEC shares. I estimate on the stacked HS6–year–end-use panel with HS6 and year × end-use fixed effects; weights are pre-period end-use share multiplied by the HS6 code's share of imports from China to the United States, and standard errors are clustered by HS6.

Table S4: Transshipment Response to Tariffs (with HS4×Year clustering and HS6, year, and HS2×Year Fixed Effects)

	Intensive Margin				Extensive Margin			
	θ_{low}		θ_{high}		$\mathbf{1}(\theta_{\text{low}} > 0)$		$\mathbf{1}(\theta_{\text{high}} > 0)$	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta_{k,t}$	0.366***		1.34***		2.38***		2.42***	
	(0.085)		(0.178)		(0.698)		(0.689)	
$\Delta_{k,t} \times$ Cap		0.312***		1.14***		2.49**		2.52**
		(0.079)		(0.207)		(0.925)		(0.913)
$\Delta_{k,t} \times$ Con		0.258***		0.780***		1.74**		1.78**
		(0.070)		(0.186)		(0.669)		(0.663)
$\Delta_{k,t} \times$ Int		0.436***		1.63***		2.42***		2.46***
		(0.102)		(0.215)		(0.603)		(0.596)
R^2	0.41	0.42	0.51	0.52	0.58	0.58	0.58	0.58
Observations	54,912	54,912	54,912	54,912	54,912	54,912	54,912	54,912
HS6 & Year & Year×HS2 FE	✓	✓	✓	✓	✓	✓	✓	✓
Weighted	✓	✓	✓	✓	✓	✓	✓	✓

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the coefficients for regressions of (12). Each HS6–year observation is fractionally allocated across end-uses according to its UN-BEC shares. I estimate on the stacked HS6–year–end-use panel with HS6, year, and year × HS2 fixed effects; weights are pre-period end-use share (since I are stacking by end-use, this is the same as unweighted), and standard errors are clustered by HS4 and year.

Table S5: Pre-Period Test

	Intensive Margin		Extensive Margin	
	θ_{low}	θ_{high}	$\mathbf{1}(\theta_{\text{low}} > 0)$	$\mathbf{1}(\theta_{\text{high}} > 0)$
	(1)	(2)	(3)	(4)
$\Delta_{k,t}$	0.267***	0.930***	1.92***	1.98***
	(0.041)	(0.129)	(0.589)	(0.580)
$\Delta_{k,t} \times \mathbf{1}(t < 2018)$				
R^2	0.39	0.48	0.51	0.52
Observations	54,912	54,912	54,912	54,912
HS6 & Year×Use FE	✓	✓	✓	✓
Weighted	✓	✓	✓	✓

* p < 0.1, ** p < 0.05, *** p < 0.01

Notes: Each column repeats the main regression specification for both intensive and extensive margins, but controls for the pre-period with an indicator variable for pre-tariff years. Each HS6–year observation is fractionally allocated across end-uses according to its UN-BEC shares. I estimate on the stacked HS6–year–end-use panel with HS6 and year × end-use fixed effects; weights are pre-period end-use share multiplied by the HS6 code's share of imports from China to the United States, and standard errors are clustered by HS6.

Table S6: Slope-adjusted difference-in-differences

	$\tilde{\theta}_{\text{low}}$	$\tilde{\theta}_{\text{high}}$	$\mathbf{1}(\tilde{\theta}_{\text{low}} > 0)$	$\mathbf{1}(\tilde{\theta}_{\text{high}} > 0)$
	(1)	(2)	(3)	(4)
$\Delta_{k,t}$	0.251***	0.804***	1.97***	2.09***
	(0.047)	(0.188)	(0.709)	(0.703)
R^2	0.39	0.46	0.51	0.51
Observations	54,912	54,912	54,912	54,912
Weighted	✓	✓	✓	✓
HS6 & Year×Use FE	✓	✓	✓	✓

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Each outcome is first residualized by its HS6-specific linear trend estimated over 2012–2017, removing product-level pre-slopes before estimation. I then re-estimate the baseline specification with HS6 and year×end-use fixed effects, weighted by 2017 China→US import shares multiplied by pre-period end-use share and clustered by HS6. Coefficients therefore capture the effect of the tariff wedge on deviations from each product's pre-trend.

Table S7: Long-differenced Specification

	θ_{Low}	θ_{High}	$\mathbf{1}(\theta_{\text{low}} > 0)$	$\mathbf{1}(\theta_{\text{high}} > 0)$	$\mathbf{1}(\theta^{\text{Act}}_{\text{low}} > 0)$	$\mathbf{1}(\theta^{\text{Act}}_{\text{high}} > 0)$
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta_{k,\text{Post}} - \Delta_{k,\text{Pre}}$	0.211^{***}	0.729^{***}	1.40^{***}	1.50^{***}	1.44^{**}	1.48^{**}
	(0.050)	(0.143)	(0.490)	(0.474)	(0.615)	(0.591)
R^2	0.15	0.24	0.10	0.11	0.15	0.15
Observations	4,246	4,246	4,246	4,246	2,640	2,513
Weighted	✓	✓	✓	✓	✓	✓
Use FE	✓	✓	✓	✓	✓	✓

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: For each HS6 I compute pre (2012–2017) and post (2018–2023) means of the outcome and the effective wedge, form long differences, and regress the long-differenced outcomes on the long-differenced wedge with end-use fixed effects. All regressions are weighted by 2017 import shares and standard errors are clustered by HS6. Columns (5)-(6) estimate an "entry" of the extensive margin in which I restrict the sample to HS6 codes with zero evidence of pre-period transshipment.

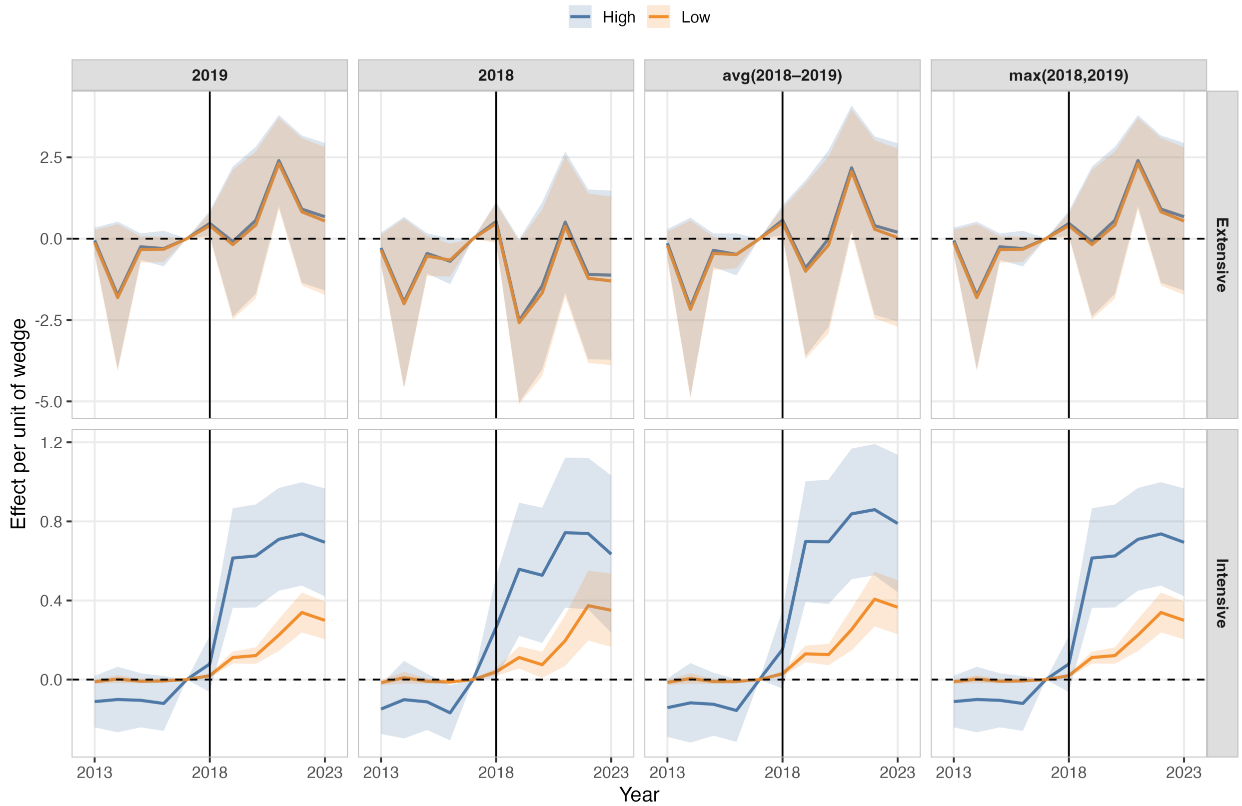
Table S8: Baseline Specification with HS4 Trends

	θ_{low}	θ_{high}	$\mathbf{1}(\theta_{\text{low}} > 0)$	$\mathbf{1}(\theta_{\text{high}} > 0)$
	(1)	(2)	(3)	(4)
$\Delta_{k,t}$	0.180^{***}	0.841^{***}	2.74^{***}	2.76^{***}
	(0.030)	(0.138)	(0.860)	(0.862)
R^2	0.54	0.61	0.59	0.59
Observations	54,912	54,912	54,912	54,912
HS6 & Year×Use FE	✓	✓	✓	✓
Weighted	✓	✓	✓	✓
HS4 Trend	✓	✓	✓	✓

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table repeats the baseline specification but adds linear trends for HS4 codes. Each HS6–year observation is fractionally allocated across end-uses according to its UN-BEC shares. I estimate on the stacked HS6–year–end-use panel with HS6 and year × end-use fixed effects; weights are pre-period end-use share multiplied by the HS6 code's share of imports from China to the United States, and standard errors are clustered by HS6.

Figure S1: Robustness to Anchoring Differently



Note: This figure estimates the event–study under four alternative definitions of the exposure anchor used to scale coefficients: (i) the 2019 effective wedge (baseline), (ii) the 2018 wedge, (iii) the average of the 2018–2019 wedges, and (iv) the maximum of the two. The dynamic profiles are nearly identical across anchors, with flat pre–trends through 2017, a sharp rise in 2018–2019, and persistent positive coefficients thereafter. The only exception is the *extensive* margin when exposure is anchored to 2018, where the coefficients appear attenuated or slightly negative immediately after 2018.

This discrepancy is mechanical. The 2018 wedge captures only the partial–year effect of tariff exposure, since most Section 301 tariffs were announced in the spring and implemented between July and September 2018. The exposure variable therefore understates the full shock in that year, while the dependent variable—the share of HS6 products exhibiting positive transshipment—responds to the cumulative policy change by the end of the year. In other words, the denominator (the wedge) is too small relative to the behavioral response, biasing the per–unit effect downward. When exposure is defined using the 2019 wedge, or alternatively using the average or maximum of the 2018–2019 wedges to account for the phase–in of tariffs, the extensive–margin coefficients align closely with the baseline pattern. The intensive–margin results are virtually unaffected across all anchors. Overall, the event–study evidence is robust to how the exposure anchor is defined, and the muted 2018–anchor response is fully consistent with the mid–year timing of tariff implementation and the partial measurement of exposure in that year.

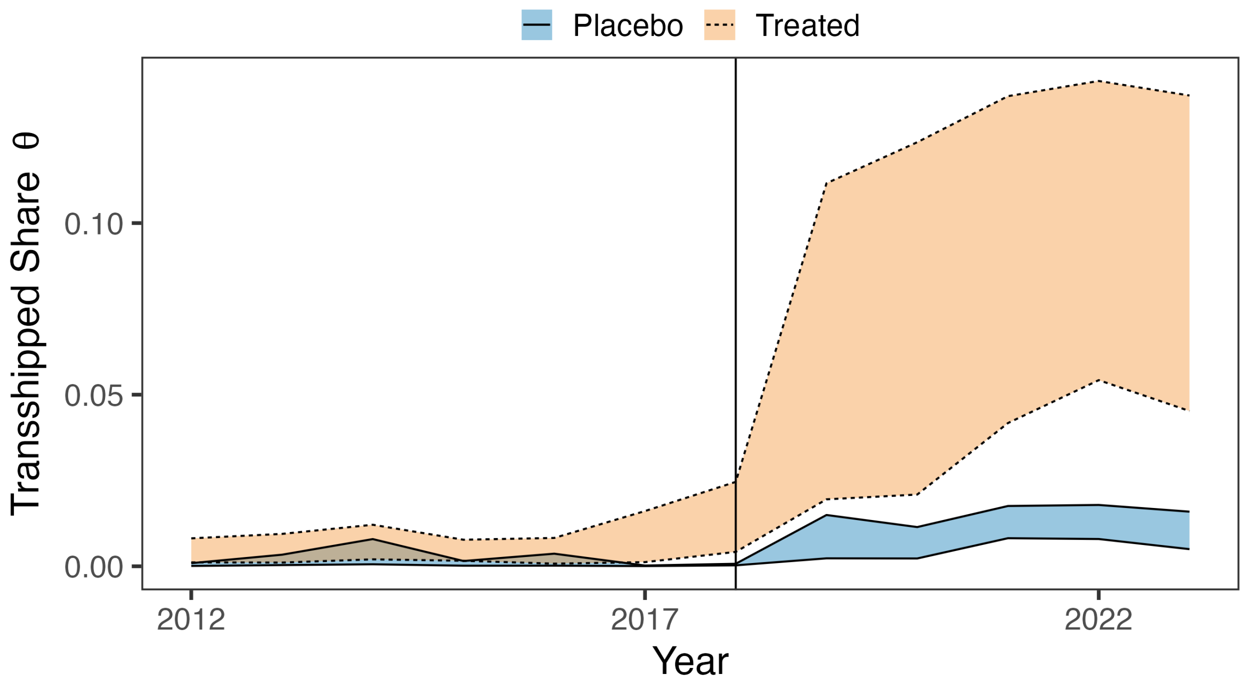
Table S9: Transshipment Response to Tariffs (2016-2019)

	Intensive Margin				Extensive Margin			
	θ_{low}		θ_{high}		$\mathbf{1}(\theta_{\text{low}} > 0)$		$\mathbf{1}(\theta_{\text{high}} > 0)$	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta_{k,t}$	0.131^{***}		0.816^{***}		1.36		1.40	
	(0.022)		(0.161)		(1.10)		(1.10)	
$\Delta_{k,t} \times$ Cap		0.116^{***}		0.777^{***}		1.15		1.14
		(0.032)		(0.248)		(1.32)		(1.32)
$\Delta_{k,t} \times$ Con		0.065^{***}		0.273^{**}		0.232		0.309
		(0.019)		(0.121)		(1.45)		(1.45)
$\Delta_{k,t} \times$ Int		0.267^{***}		1.75^{***}		3.64^{***}		3.74^{***}
		(0.041)		(0.267)		(0.481)		(0.490)
R^2	0.33	0.33	0.43	0.44	0.47	0.48	0.46	0.47
Observations	22,880	22,880	22,880	22,880	22,880	22,880	22,880	22,880
HS6 & Year×Use FE	✓	✓	✓	✓	✓	✓	✓	✓
Weighted	✓	✓	✓	✓	✓	✓	✓	✓

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the coefficients for regressions of (12). Each HS6–year observation is fractionally allocated across end-uses according to its UN-BEC shares. I estimate on the stacked HS6–year–end-use panel with HS6 and year × end-use fixed effects; weights are pre-period end-use share multiplied by the HS6 code's share of imports from China to the United States, and standard errors are clustered by HS6.

Figure S2: Aggregate Transshipment Comparing Treated to Untreated Codes



Notes: This shows the intensive margin of transshipment for both treated and untreated HS6 codes following the construction as described in the text.

Table S10: The Entropy-Weighted Effect of Tariffs on Transshipment

	Intensive Margin		Extensive Margin	
	θ_{low}	θ_{high}	$\mathbf{1}(\theta_{\text{low}} > 0)$	$\mathbf{1}(\theta_{\text{high}} > 0)$
	(1)	(2)	(3)	(4)
$\Delta_{k,t}$	0.262^{***}	0.909^{***}	2.01^{***}	2.06^{***}
	(0.045)	(0.142)	(0.538)	(0.527)
R^2	0.39	0.48	0.51	0.52
Observations	54,912	54,912	54,912	54,912
HS6 & Year×Use FE	✓	✓	✓	✓
Weighted	✓	✓	✓	✓

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

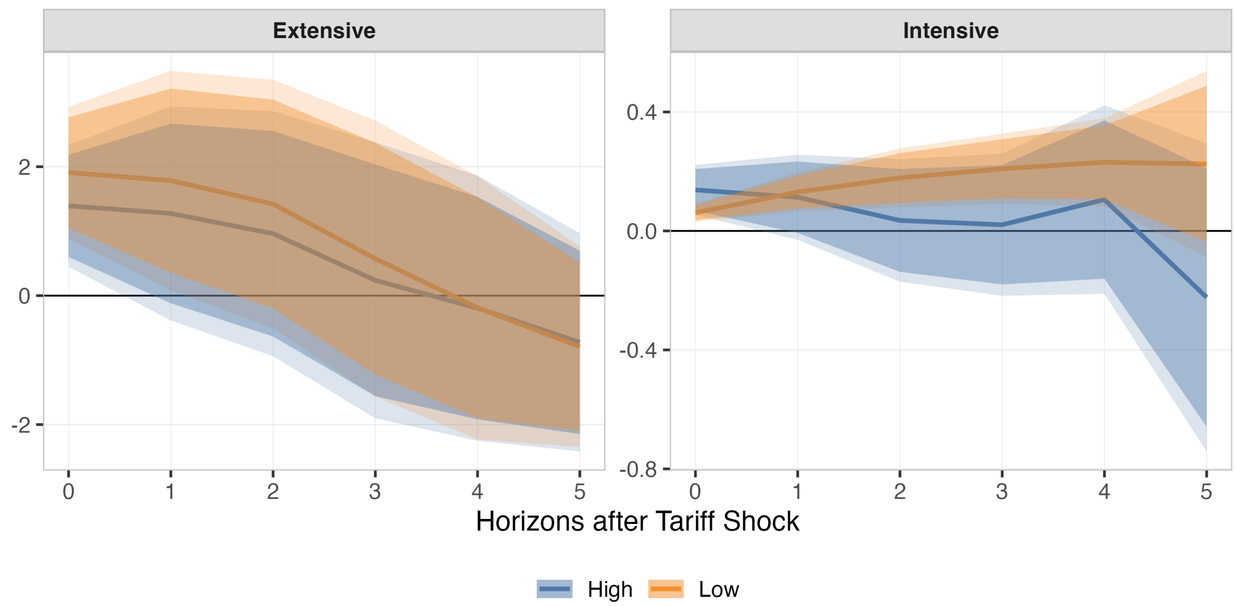
Notes: I re-weight HS6 units via entropy balancing so that treated (top quartile of pre-period wedge) and control have identical pre-period moments of diversion (winsorized and standardized). The panel regressions are re-estimated with weights = 2017 exposure × EB weights × pre-period end-use share, HS6 and year×end-use fixed effects, and HS4-clustered SEs. Coefficients are per unit of the wedge. Estimates are effectively unchanged, indicating the results are not driven by pre-period imbalances.

Table S11: Percent difference between baseline and leave-one-out: $100 \times \frac{\beta_{LOO}-\beta}{\beta}$

Country Code	Extensive (High)	Extensive (Low)	Intensive (High)	Intensive (Low)
CAN	-17.741	-18.909	18.108	19.409
MEX	4.916	4.330	32.779	24.288
DEU	-8.019	-7.600	23.814	15.804
JPN	-3.000	-4.312	31.010	29.929
TWN	-10.480	-10.849	3.412	7.809
SGP	-17.408	-16.475	2.298	3.057
MYS	-1.852	-1.249	3.504	10.467
VNM	-45.038	-45.398	0.107	4.097
HKG	-13.705	-14.733	0.444	2.080
THA	-1.719	-1.088	3.618	6.750
RUS	-16.286	-17.559	0.602	-0.242
IDN	-1.840	-1.431	1.912	5.003
POL	-1.388	-1.840	2.306	2.079
TUR	1.561	1.726	1.426	1.654
KHM	-1.019	0.833	1.190	6.889

Notes: For each regression specification, I leave out one hub and recompute the coefficient. After that, I compute the percent difference between the leave-one-out estimate and the baseline regression. For example, the leave-one-out coefficient when excluding Canada from the sample is 17 percent smaller than the baseline intensive margin coefficient from Column 1 of Table C.2. The table shows the most influential fifteen hubs.

Figure S3: Local Projections of Transshipment from Tariffs



Note: This figure estimates a local projection of the form

$$f(\theta_{k,t+h}) - f(\theta_{k,t-1}) = \alpha_i + \delta_{t\times e(k)} + \psi_{t+h}\Delta_{k,t} + \varepsilon_{k,t+h}.$$

I do this for both the extensive margin, using the linear probability model from the main text, as well as an intensive margin with the inverse hyperbolic sine transformation. Standard errors are clustered by HS6 code.

Table S12: Composition vs. Selection: Price Response to θ

	$\Delta \ln p_{\mathrm{CN}}$			
	(1)	(2)	(3)	(4)
θ_{low}	-0.276*		-0.259	
	(0.150)		(0.162)	
θ_{high}		-0.121**		-0.115**
		(0.053)		(0.050)
$\theta_{\mathrm{low}} \times$ UV dispersion			-0.028	
			(0.035)	
$\theta_{\mathrm{high}} \times$ UV dispersion				-0.027
				(0.031)
R^2	0.09	0.09	0.09	0.09
Observations	46,474	46,474	46,474	46,474
HS6 & Year×Use FE	✓	✓	✓	✓
Weighted	✓	✓	✓	✓

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Regressions of the change in China unit values on θ (columns 1–2) and on θ interacted with the standardized pre-tariff within-HS6 unit-value dispersion (columns 3–4). A non-zero interaction suggests selection (high-dispersion HS6 codes have different price responses), while a flat baseline suggests pure composition. All specifications include HS6 and year×end-use fixed effects, weight by 2017 China→US exposure times pre-period end-use share, and cluster standard errors by HS6.